



北京大学
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A Novel Ferroelectric Charge-Steering Memcapacitor Array based Δ -CIM with Enhanced Parallelism and Self-Adaptive Sparsity for Edge LLMs

Jin Luo^{#*}, Shaodi Xu[#], Peiyu Chen, Yuxin Lin, Haoyu Wang, Bingrui Song,
Weikai Xu, Boyi Fu, Yuan Wang, Le Ye, Qianqian Huang^{*}, Ru Huang^{*}

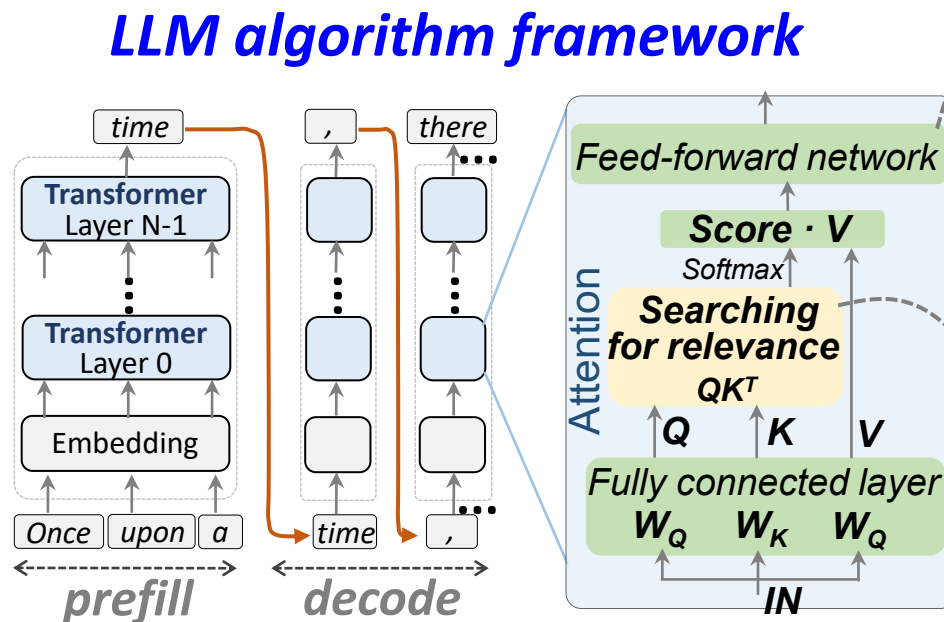
[#]Equal contribution

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Motivation (1/5)

Large language models (LLMs)

- Computational foundation for hyperscale large language models (LLMs)
 - High-density storage for parameters & massive computing for execution
 - Dominant operations: **multiply-accumulate (MAC)** & *Key-Value cache searching*



MAC operation

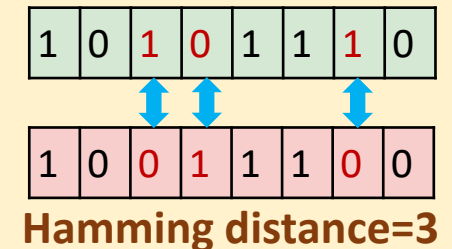
$$Out_j = \sum_i IN_i \times w_{ij} \quad [x_1, \dots, x_m] \cdot \begin{bmatrix} w_{11} & \dots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{m1} & \dots & w_{mn} \end{bmatrix}$$

Partial sum

Searching operation

$$Out_{Hamming} = \sum_i q_i \oplus k_{ij}$$

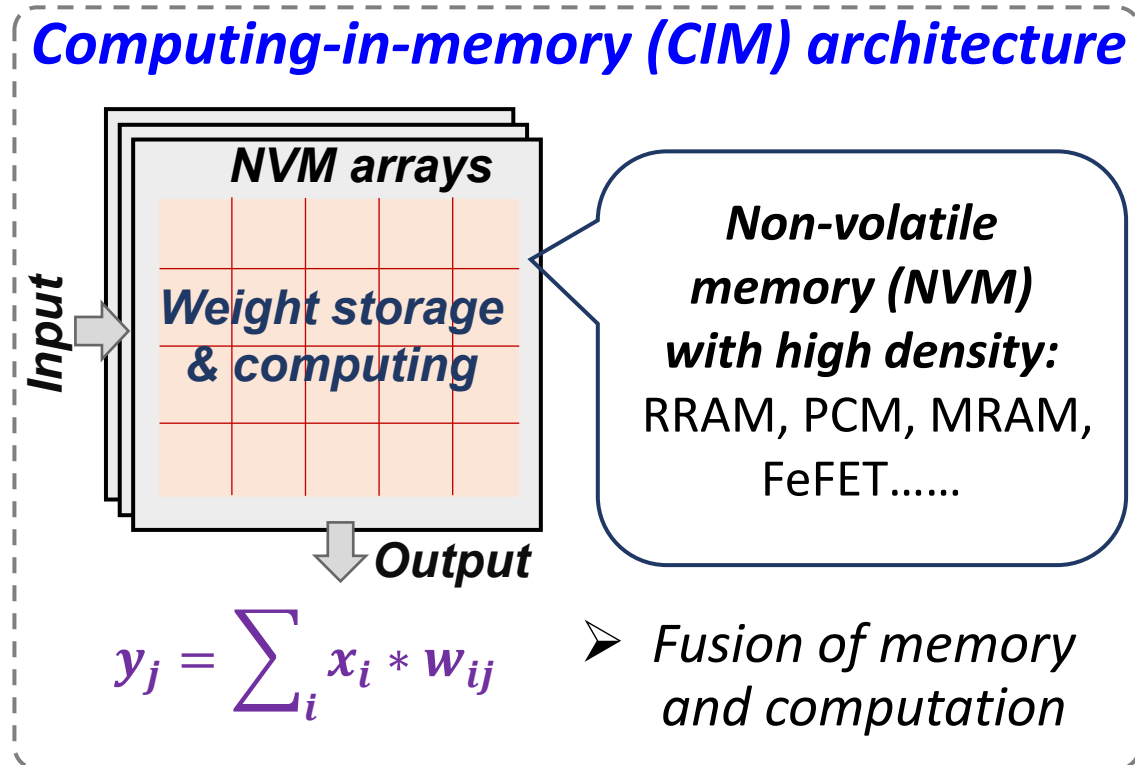
Similarity for search



Motivation (2/5)

Computing-in-memory (CIM) for LLMs

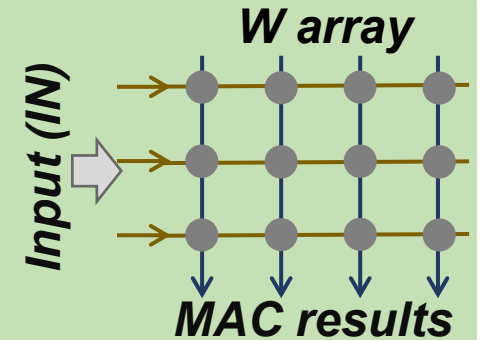
- Computing-in-memory (CIM) architecture for LLMs
 - Highly-parallel MAC** and **searching operations** inside memory array
 - High compute density and energy efficiency with further 3D-stackable capacity



MAC operation in CIM

$$Out_j = \sum_i IN_i \times w_{ij}$$

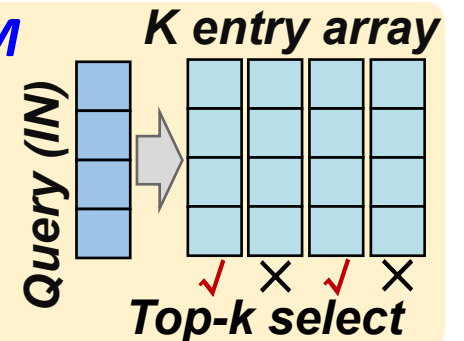
Partial sum



Searching operation in CIM

$$Out_{Hamming} = \sum_i q_i \oplus k_{ij}$$

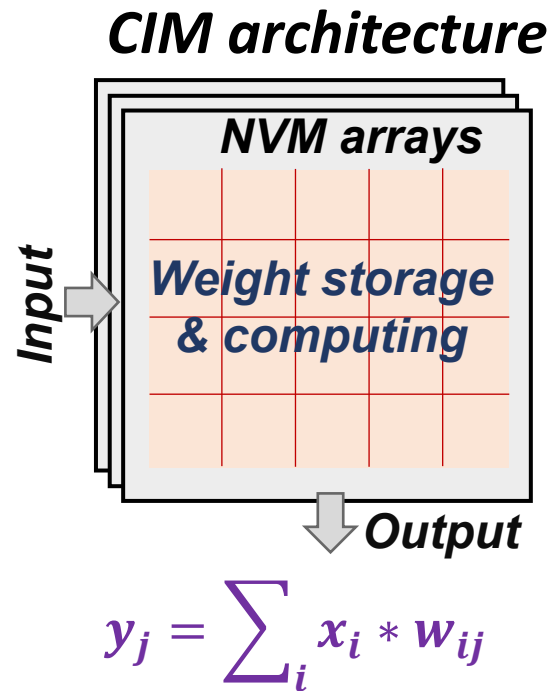
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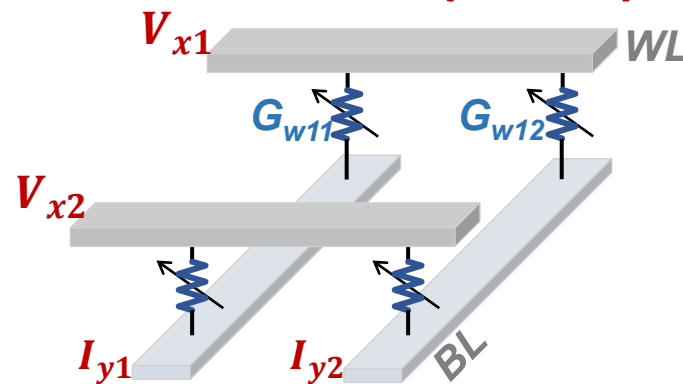
Motivation (3/5)

Capacitive computing-in-memory (C-CIM)

- Emerging capacitive NVM based CIM (C-CIM)
 - Programmable capacitance states** → weight storage & charge-domain computing
 - Eliminate direct conduction (DC) path → **mitigate thermal dissipation**



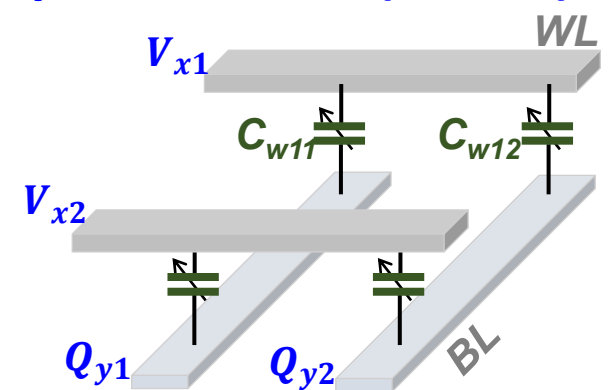
Resistive CIM (R-CIM)



Current domain: $I_{sum} = \sum V_x \cdot G_w$

- High static current and static power
- Inevitable thermal dissipation

Capacitive CIM (C-CIM)



Charge domain: $Q_{sum} = \sum V_x \cdot C_w$

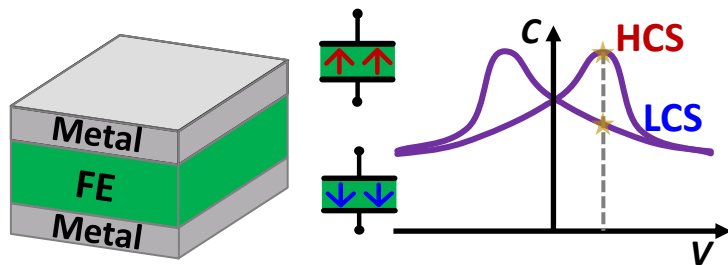
- ✓ Immune to IR drop and sneak paths
- ✓ Save static power consumption

Motivation (4/5)

Memcapacitor for C-CIM

- Memcapacitor: capacitive NVM for C-CIM
 - Ferroelectric (FE) capacitor: limited capacitance memory window (C_{MW})
 - **Programmable MOS capacitor: Improve C_{MW} due to integrated depletion layer**

Ferroelectric (FE) capacitor

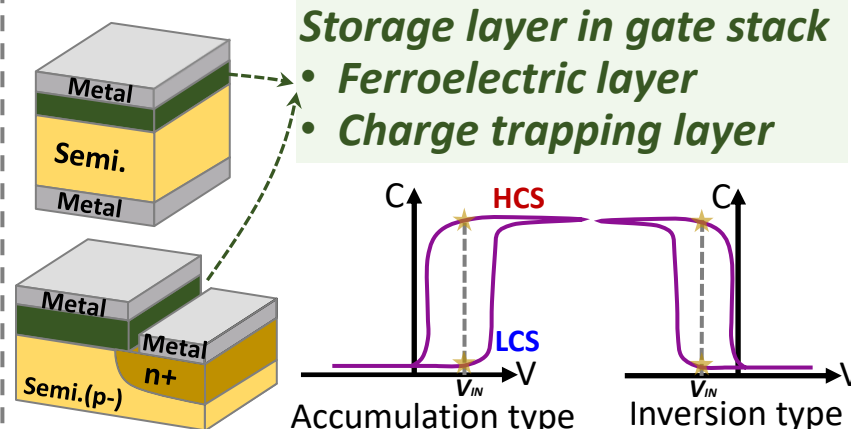


Metal-FE-metal (MFM) structure

- Mechanism: FE domain wall creep

S. Mukherjee, et al., IEDM 2023
Jin Luo, et al., IEDM 2024

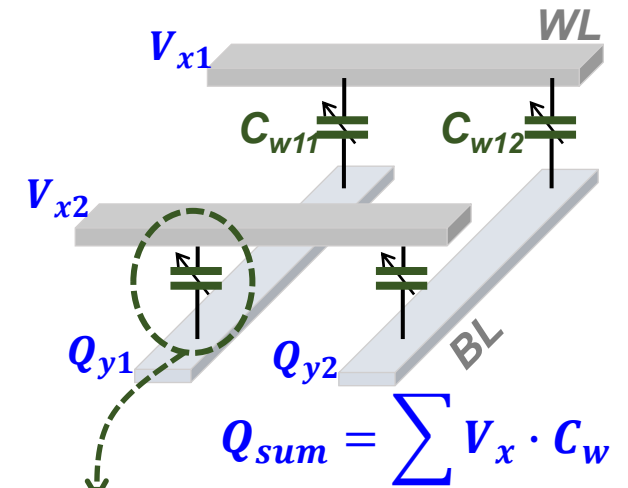
Programmable MOS capacitor



HCS:

LCS:

Capacitive CIM (C-CIM)



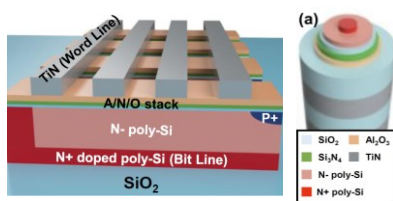
Memcapacitor:
programmable capacitor

Motivation (5/5)

Memcapacitor based C-CIM scheme

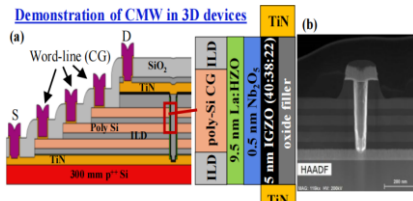
- **Single-ended** weight storage and computation in analog domain
 - Limit signal-to-noise ratio (SNR) → **Compromise parallelism and throughput**
- **Per-cycle charging input scheme** conditioned on absolute input bit value
 - Redundant dynamic power for bit stream → **Degrade energy efficiency**

Memcapacitor for C-CIM



Charge-trap modulated MOS Cap

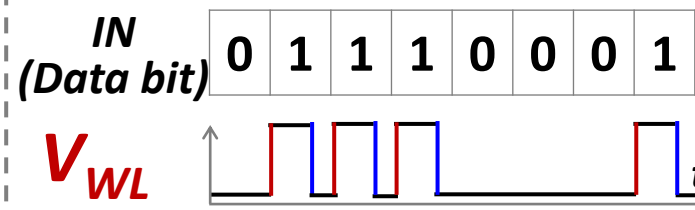
J. Yu, et al., IEDM, 2024.
M. Song, et al., IEDM, 2023



FE modulated MOS Cap.

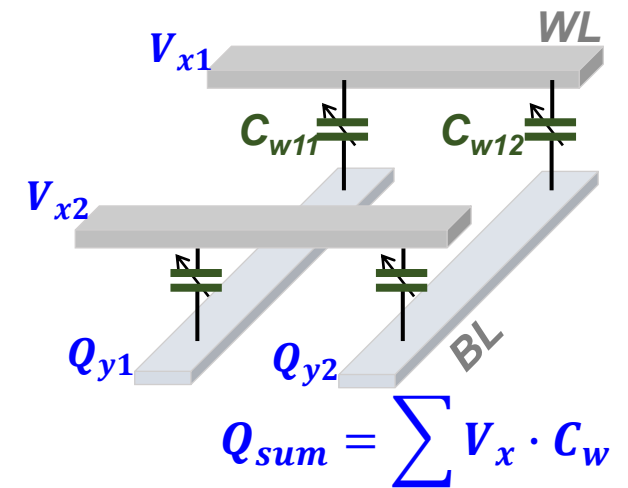
S. Mukherjee, et al., IEDM, 2024.
Z. Zhou, et al., VLSI, 2022.

Input scheme for C-CIM



- Charge and discharge word-lines even for successive identical bit '1'

Pre-cycle charging
→ High dynamic power



☹️ The two-terminal structure only supports single-ended operation
→ Parallelism/ throughput constraint

In this work of capacitive CIM for LLMs

- A novel ferroelectric (FE) charge-steering memcapacitor (FS-Cap) based Δ -CIM with **spatial and temporal delta-encoding**, enhancing parallelism and energy efficiency

FS-Cap device for spatial ΔC storage

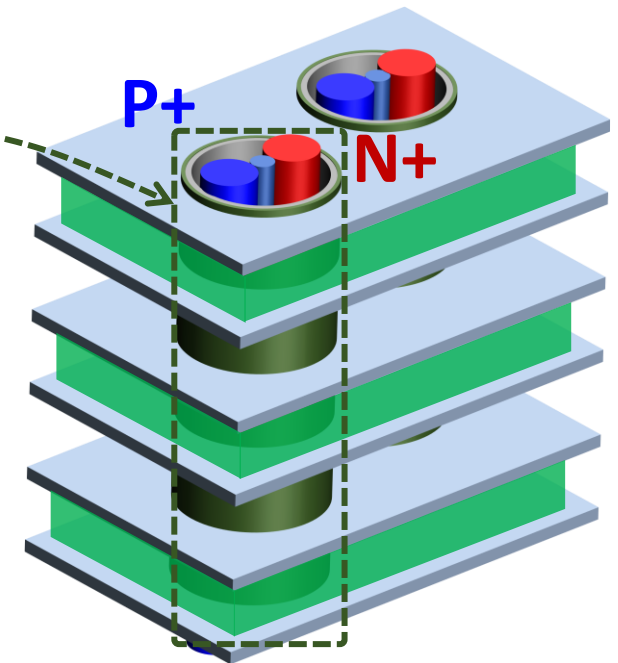
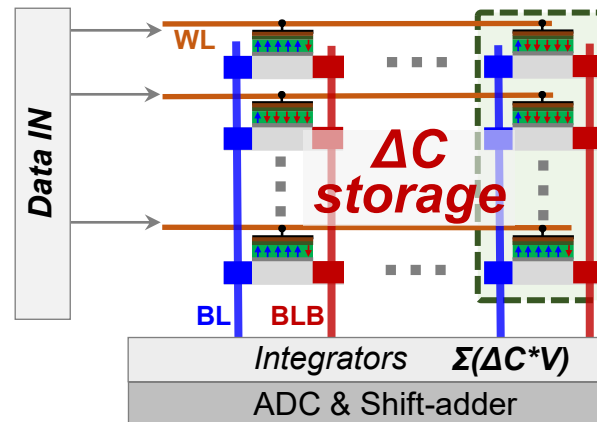
Asymmetric doping

Differential dual-ended extraction of MOS structure

FE-assisted gate-injection

Steer induced-charge between dual-ends

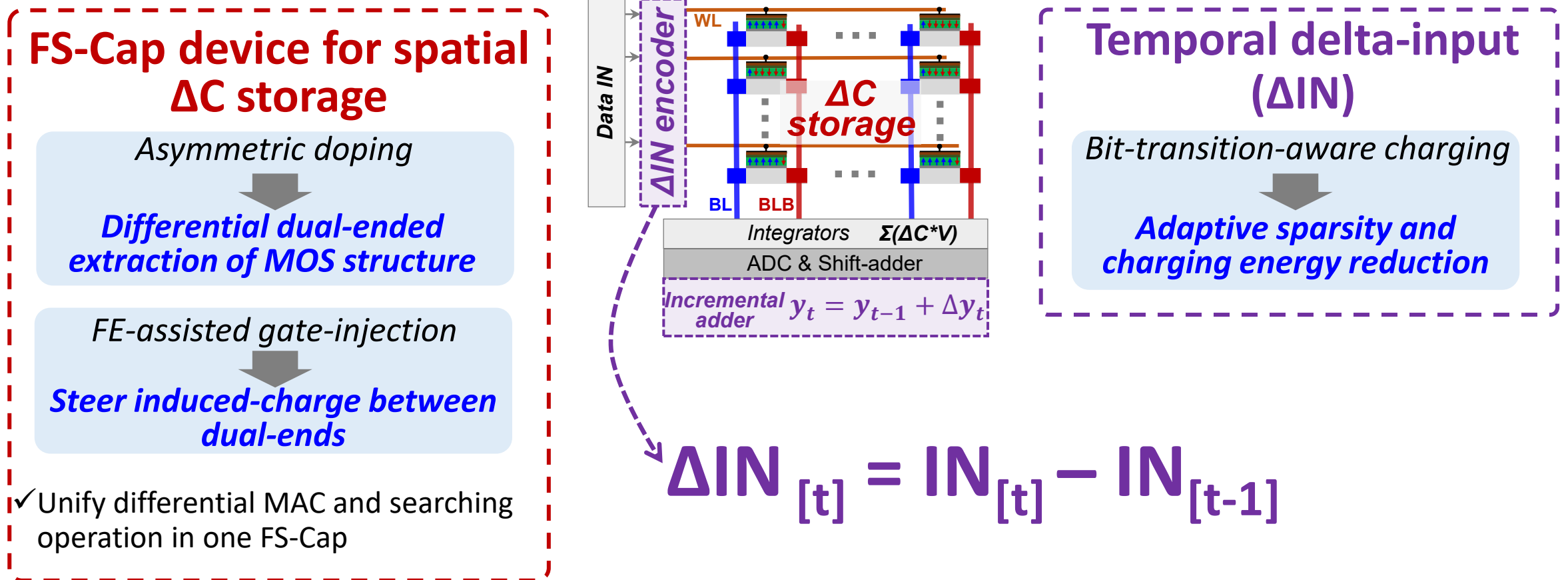
- ✓ Unify differential MAC and searching operation in one FS-Cap



AND-type 3D integration of FS-Cap structure

In this work of capacitive CIM for LLMs

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FS-Cap device for spatial ΔC storage

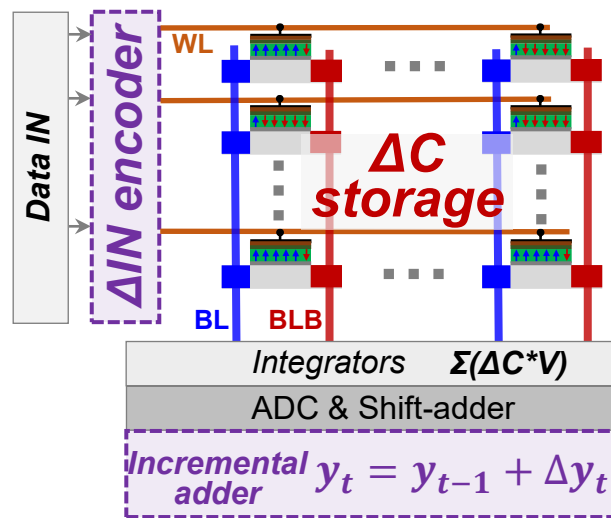
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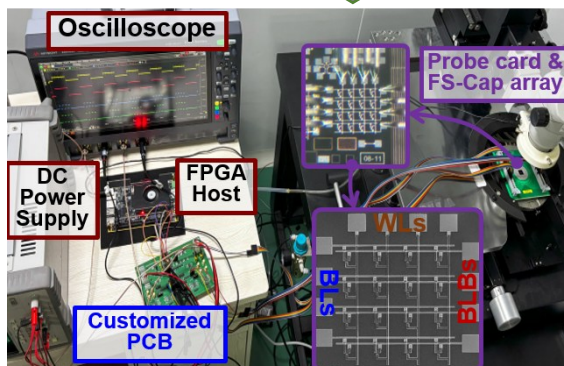
- ✓ Unify differential MAC and searching operation in one FS-Cap



Temporal delta-input (Δ IN)

Bit-transition-aware charging

Adaptive sparsity and charging energy reduction



FS-Cap-based Δ -CIM for LLMs

- ✓ Inference and fine-tuning
- ✓ **High SNR and parallelism**
- ✓ **High energy efficiency**

Outline

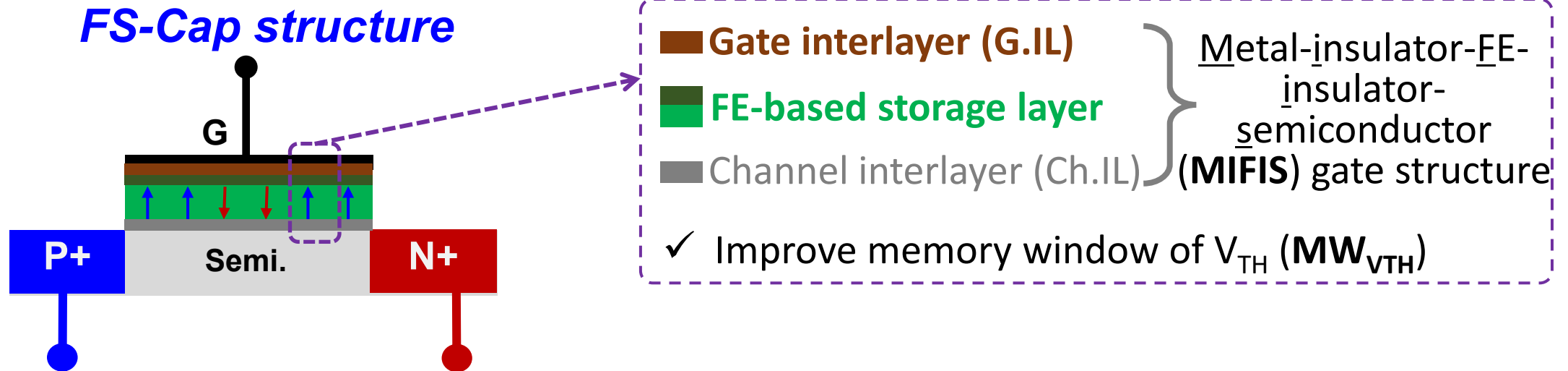
- FE charge-steering memcapacitor (FS-Cap) for Δ -CIM
- FS-Cap array based Δ -CIM behavior
- FS-Cap based Δ -CIM macro for LLMs
- Summary

Outline

- **FE charge-steering memcapapacitor (FS-Cap) for Δ -CIM**
- Design and mechanism of FS-Cap for ΔC storage
- Fabrication of FS-Cap array for Δ -CIM
- FS-Cap array based Δ -CIM behavior
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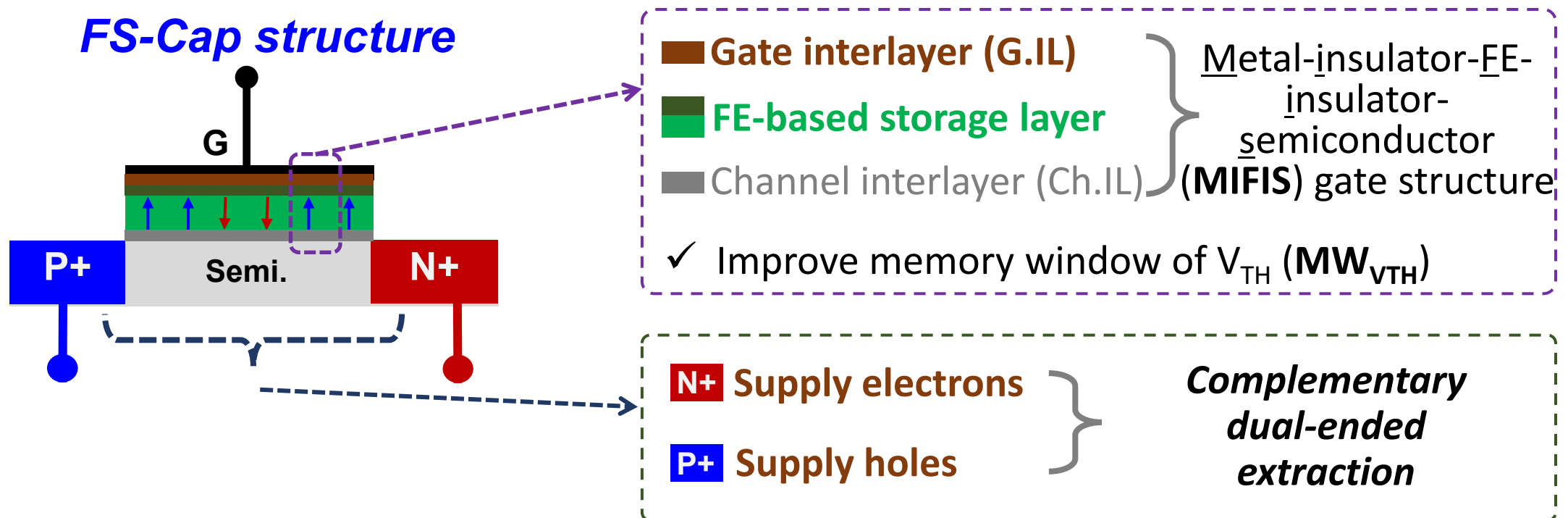
Ferroelectric charge-steering memcapacitor (FS-Cap) design

- Device structure of novel **FS-Cap**
 - Gate stack: **ferroelectric (FE) based storage layer** → modulate V_{TH}



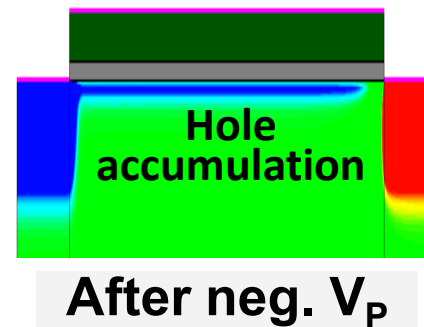
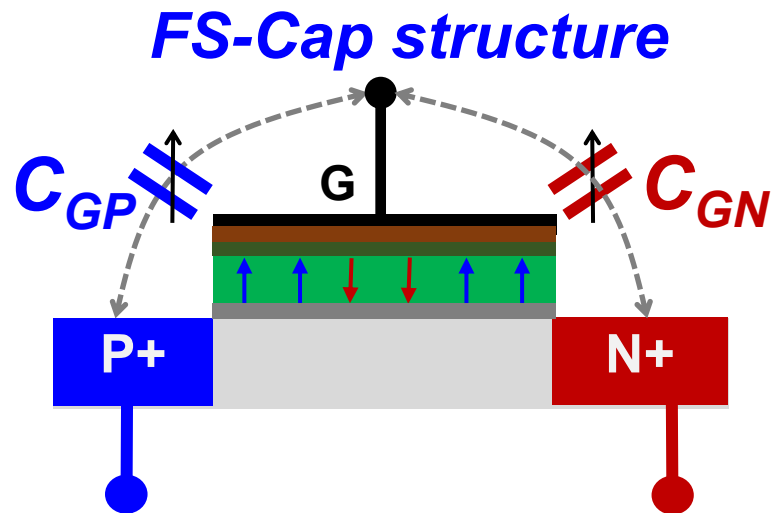
Ferroelectric charge-steering memcapacitor (FS-Cap) design

- Device structure of novel **FS-Cap**
 - Gate stack: **ferroelectric (FE) based storage layer** → modulate V_{TH}
 - **Asymmetric-doping source/drain** → supply electrons/holes to channel surface

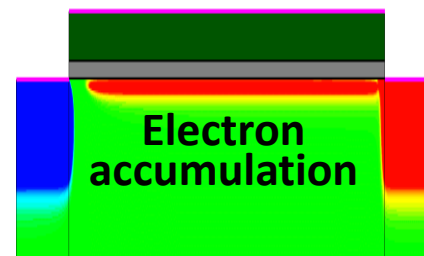


Charge-steering mechanism for ΔC in FS-Cap (1/3)

- FE-assisted charge trapping for capacitive weight storage
 - Programming voltage (V_p) to the gate $\rightarrow$ store the V_{TH} state $\rightarrow$ **switch channel carrier types** $\rightarrow$ modulate port capacitance of C_{GP} and C_{GN}

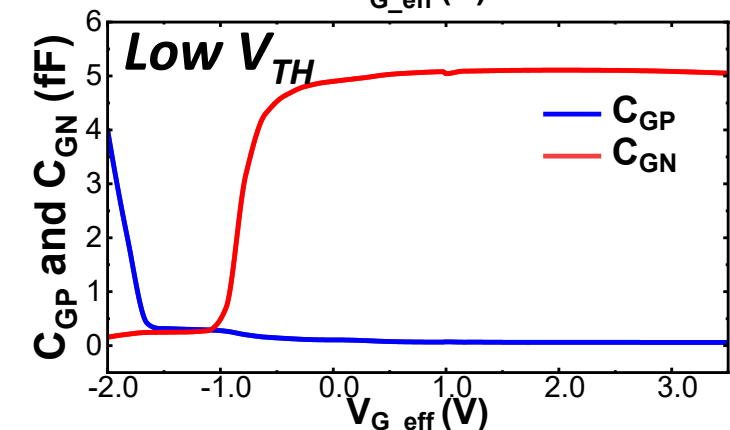
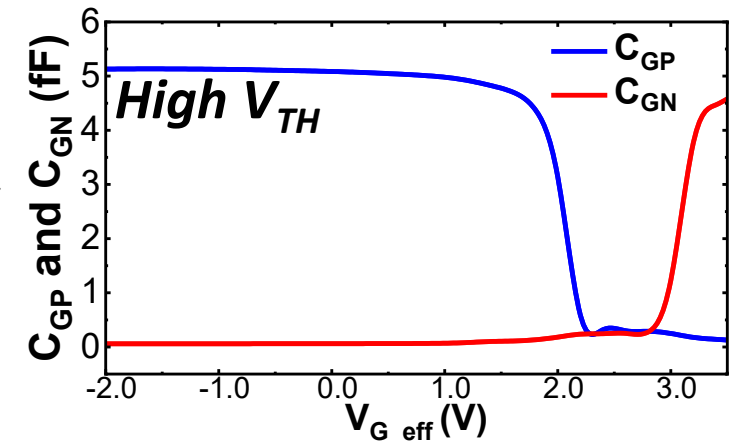


After neg. V_p



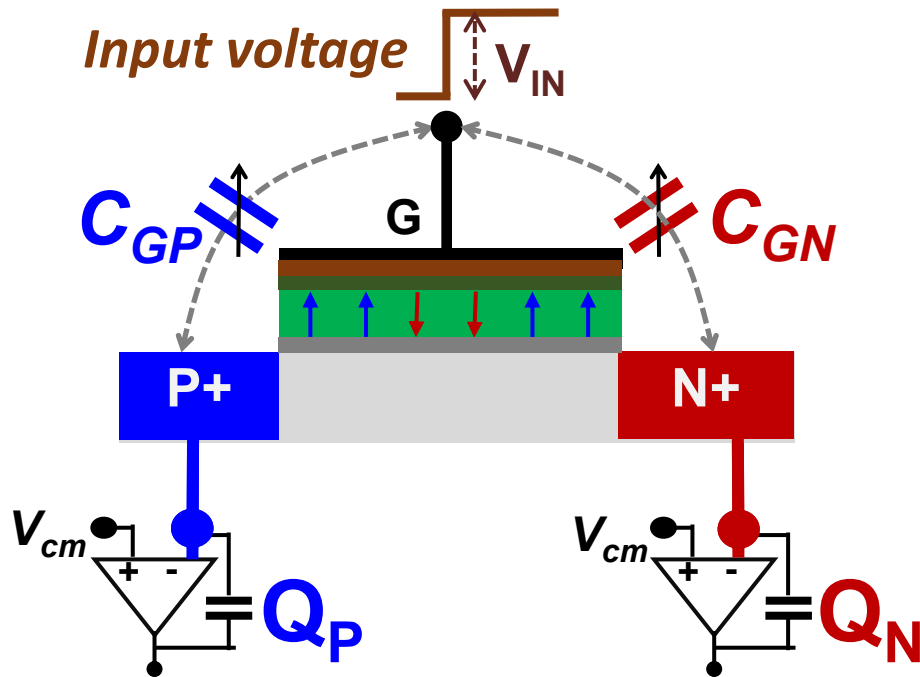
After pos. V_p

Carrier concentration (cm^{-3})
+10¹⁹ 0 -10¹⁹

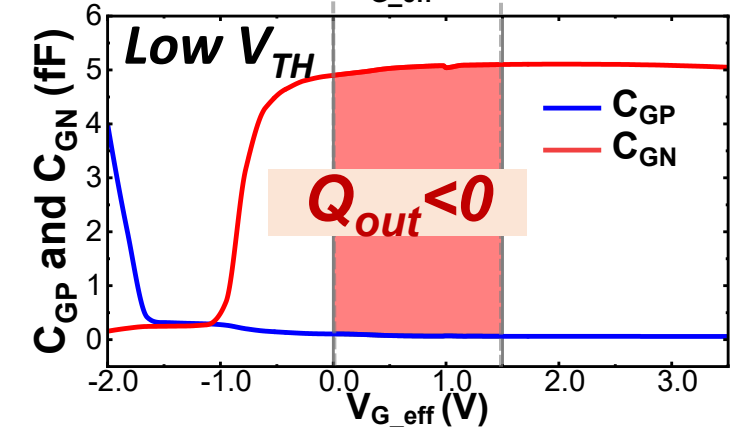
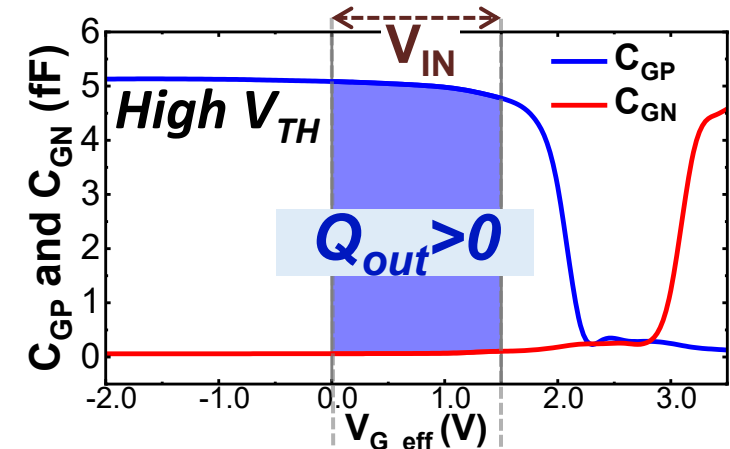
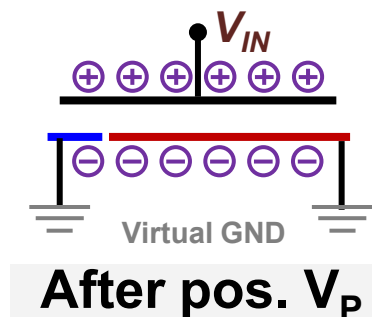
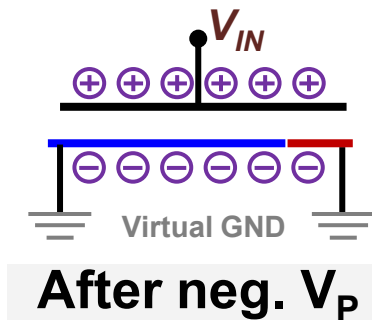


Charge-steering mechanism for ΔC in FS-Cap (2/3)

- **Charge-steering mechanism** for differential operation in FS-Cap
 - Input voltage (V_{IN}) to the gate $\rightarrow$ **induced charge is steered** to P- or N-type port dependent on C_{GP} and C_{GN} states $\rightarrow$ **differential outputs** ($Q_{out}=Q_P-Q_N$)



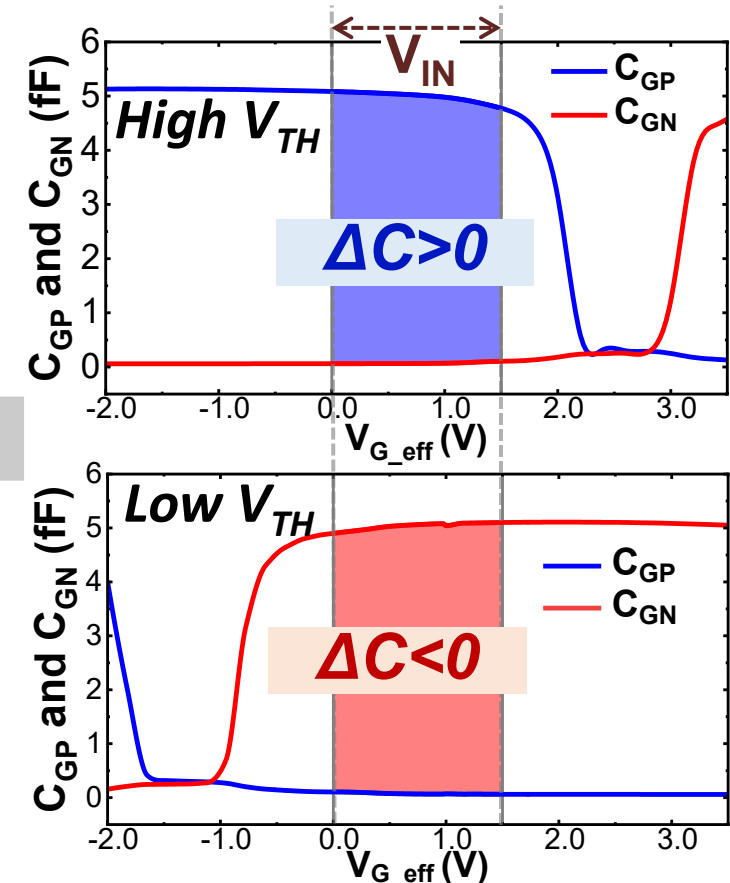
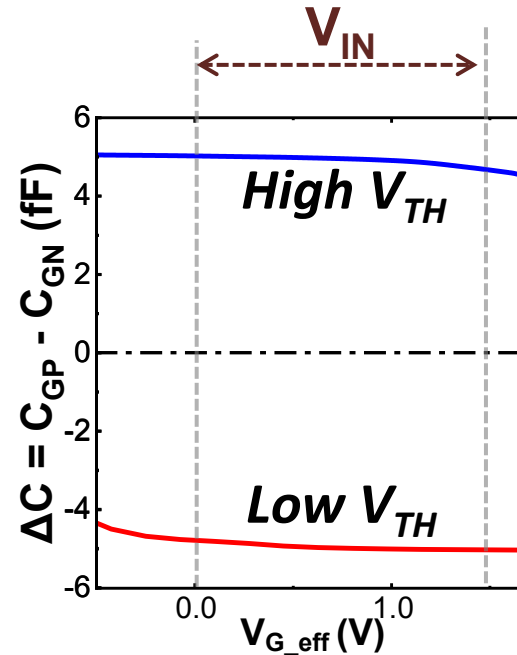
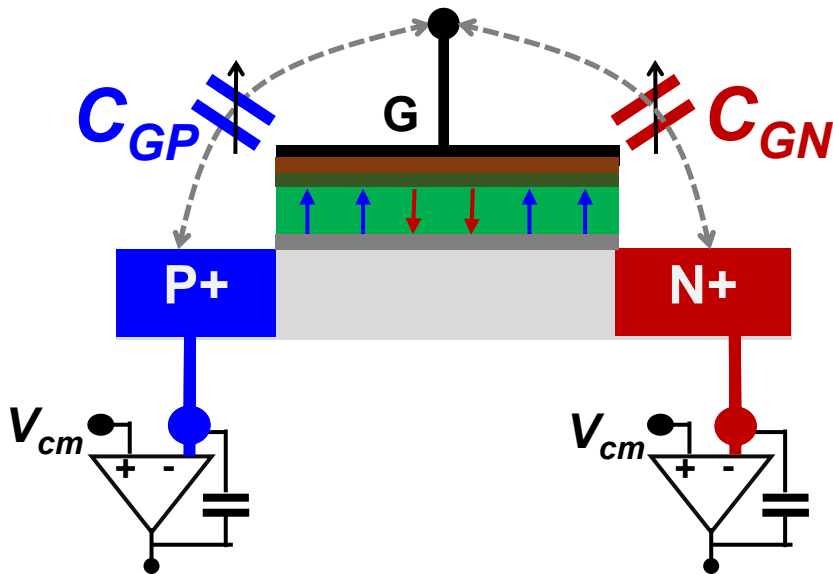
$$Q_{out} = Q_P - Q_N$$



Charge-steering mechanism for ΔC in FS-Cap (3/3)

- **Charge-steering mechanism** based ΔC storage
 - Encode weight parameters as **differential capacitance between P/N type ports**

$$\Delta C = C_{GP} - C_{GN}$$



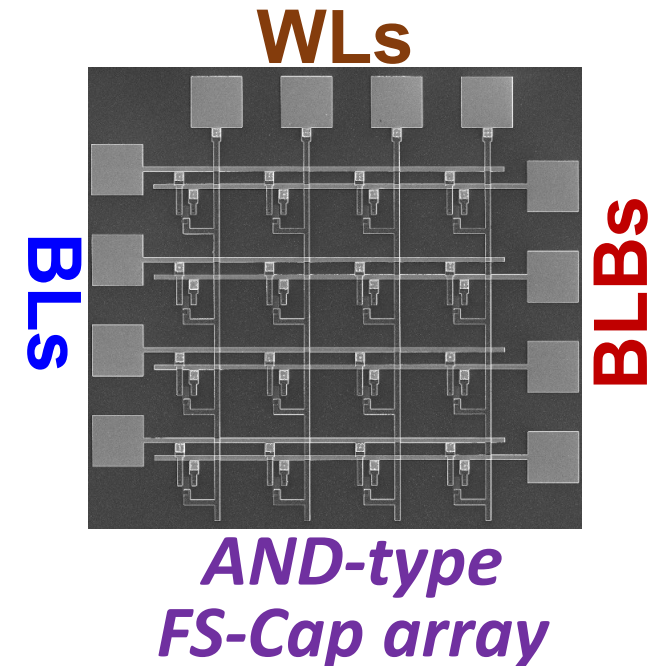
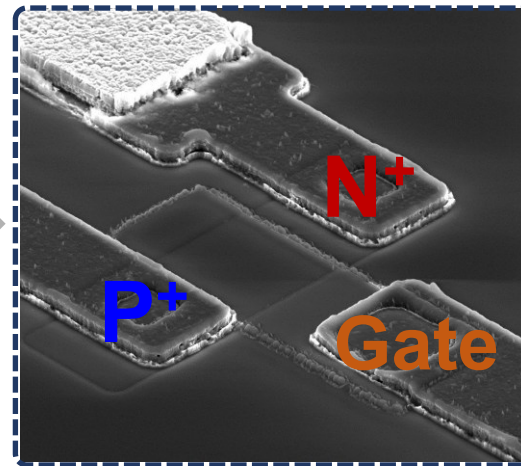
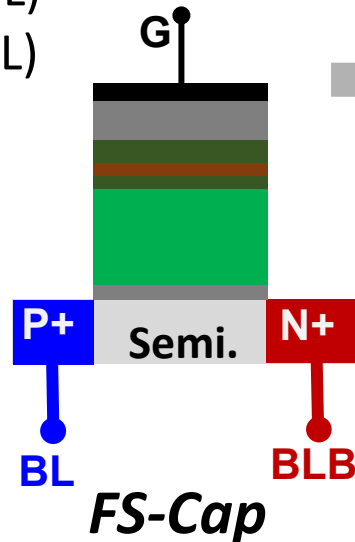
Outline

- **FE charge-steering memcapapacitor (FS-Cap) for Δ -CIM**
- Design and mechanism of FS-Cap for ΔC storage
- Fabrication of FS-Cap array for Δ -CIM
- FS-Cap array based Δ -CIM behavior
- FS-Cap based Δ -CIM macro for LLMs
- Summary

FS-Cap array fabrication (1/3)

- **Spatially separated P/N ports** for ΔC array
 - **Bipolar implantation** → carrier-specific **charge steering** to differential ports
 - Two-tier metal interconnection for FS-Cap array with **differential bit-lines (BL/BLB)**

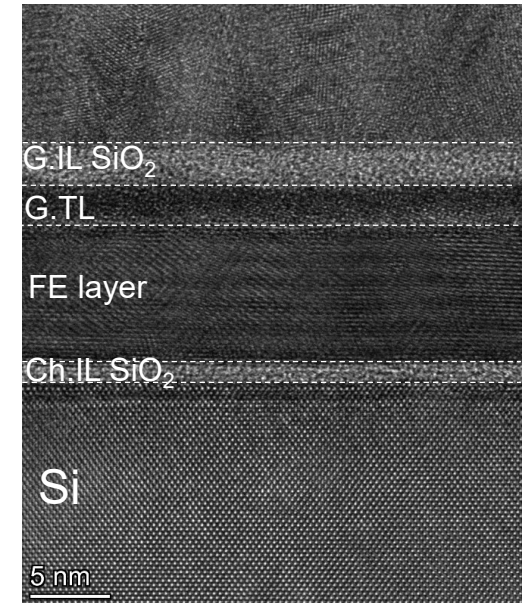
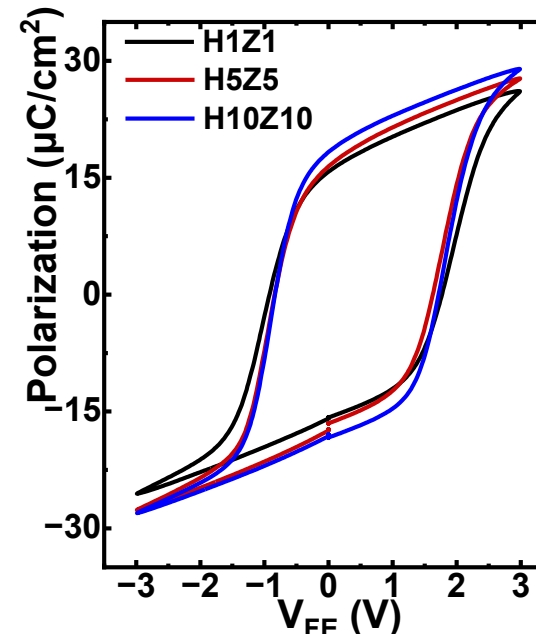
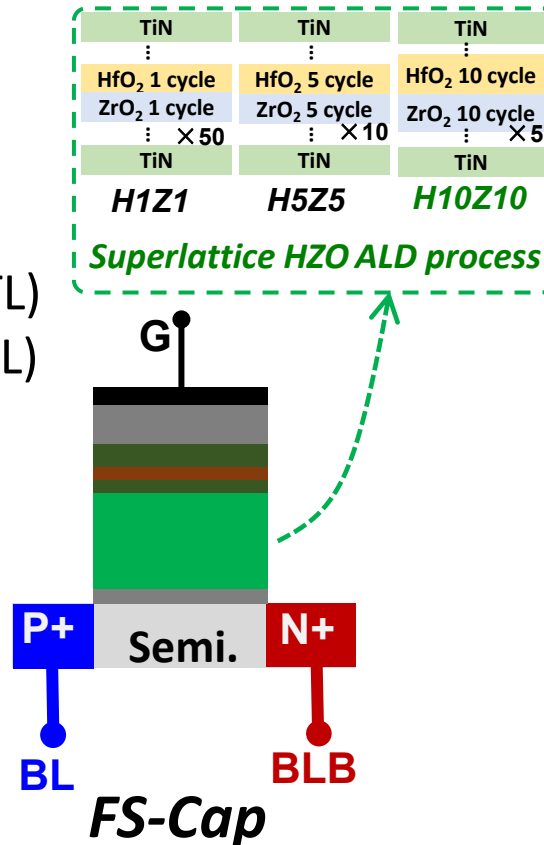
- Isolation
- Substrate clean (Ch.IL form)
- ALD FE layer, 8nm
- ALD gate-trapping layer (G.TL)
- ALD gate interface layer (G.IL)
- PVD TiN 65nm
- Gate patterning
- Ion Implantation
 - **N⁺** : As (20KeV 2E15)
 - **P⁺** : BF₂ (10KeV 2E15)
- RTP in N₂ (600°C 60s)
- ILD, Contact & metallization



FS-Cap array fabrication (2/3)

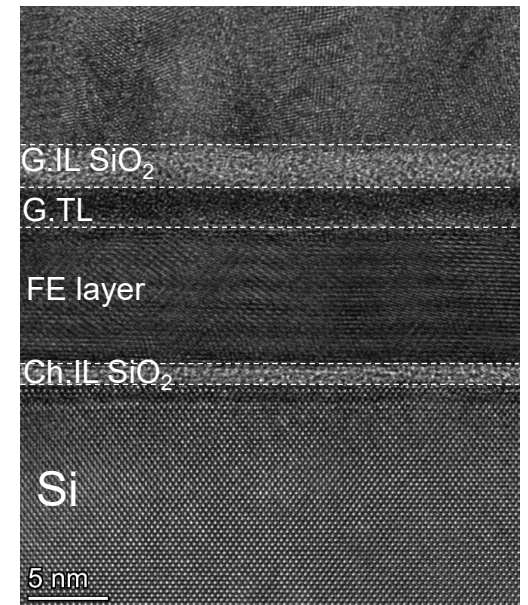
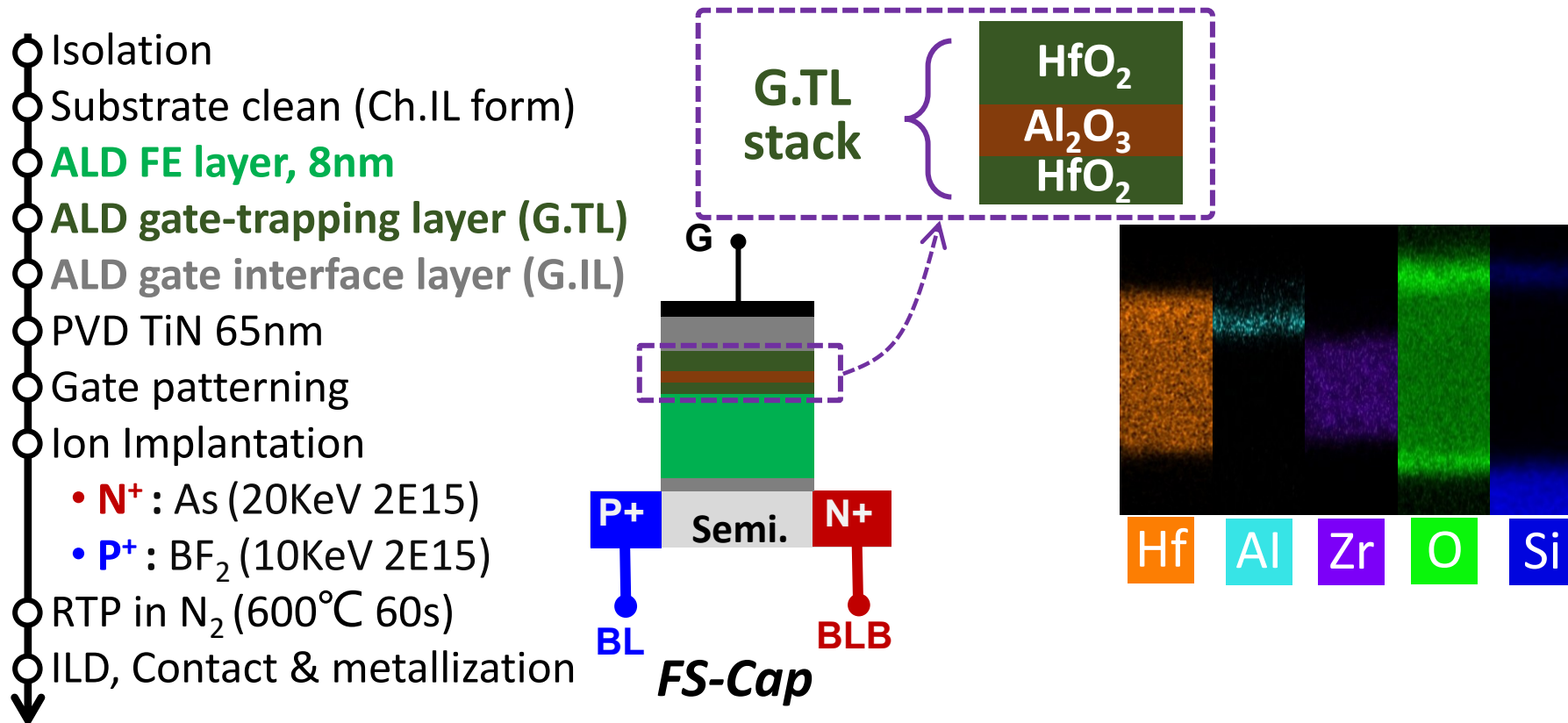
- “MIFIS” gate stack: synergize FE polarization and gate-injection
 - **FE layer**: superlattice $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2 \rightarrow$ tightened E_c distribution and high P_r

- Isolation
- Substrate clean (Ch.IL form)
- **ALD FE layer, 8nm**
- ALD gate-trapping layer (G.TL)
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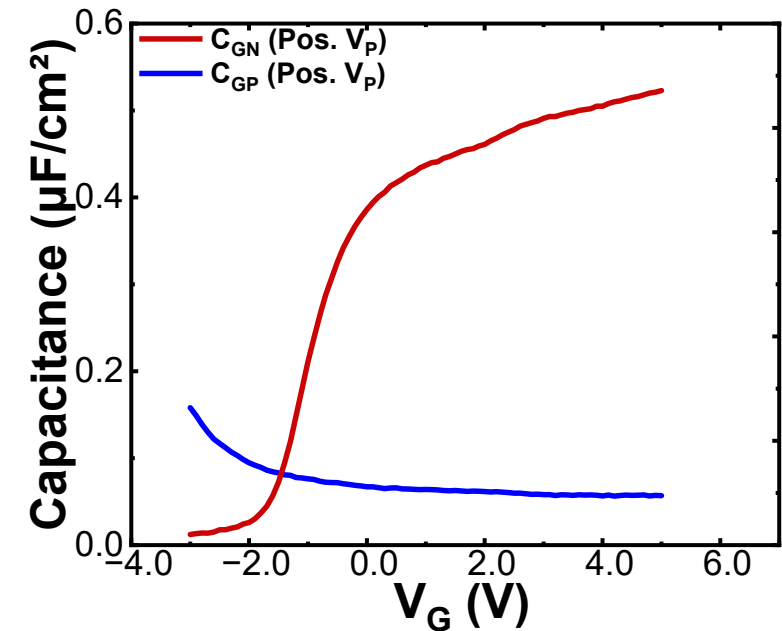
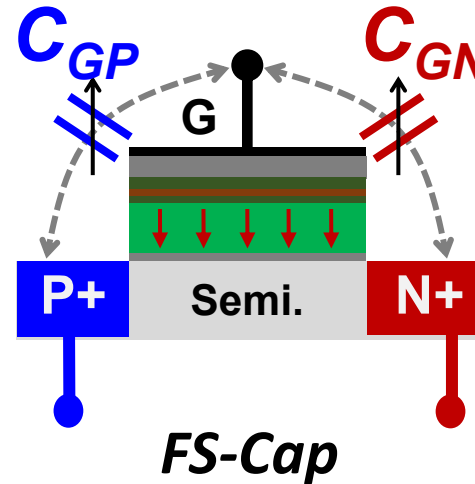
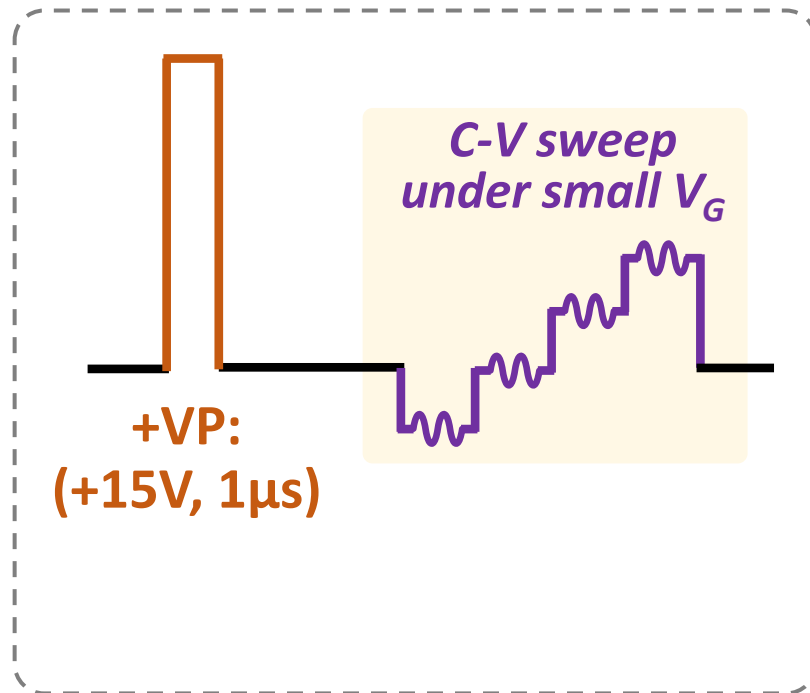
FS-Cap array fabrication (3/3)

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 - **FE layer**: superlattice $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2 \rightarrow$ tightened E_c distribution and high P_r
 - **Gate-trapping layer (G.TL)**: HfO_2 with Al_2O_3 barrier $\rightarrow$ enhance trapping efficiency



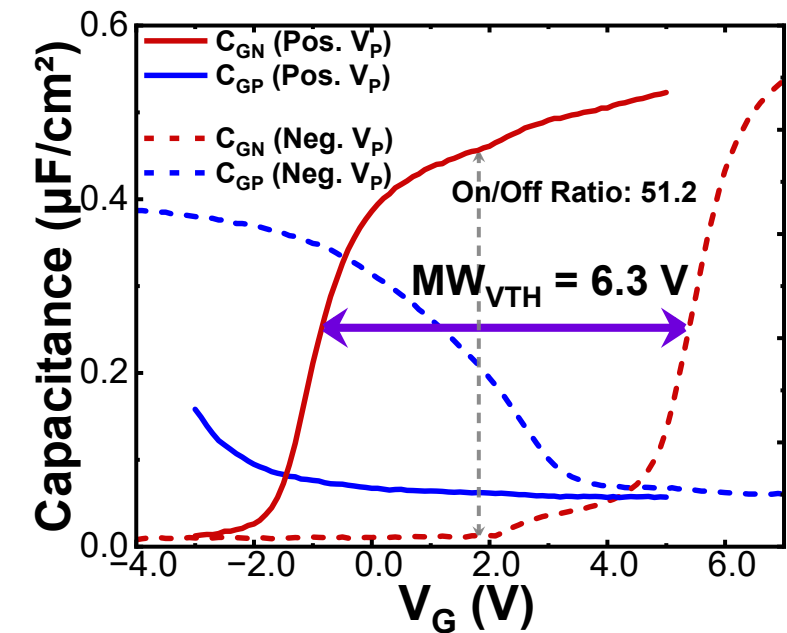
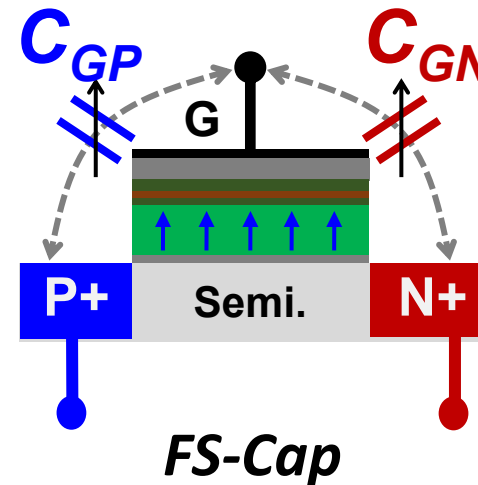
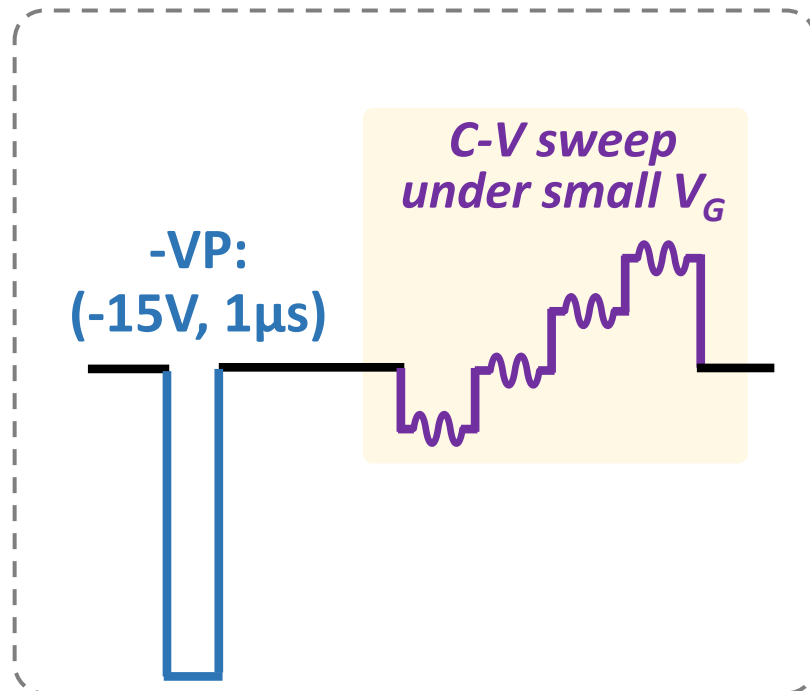
FS-Cap for ΔC storage (1/4)

- Programmed C_{GP} and C_{GN} in FS-Cap
 - Positive program voltage ($+V_P$) $\rightarrow$ **Leftward** C-V curve shift



FS-Cap for ΔC storage (2/4)

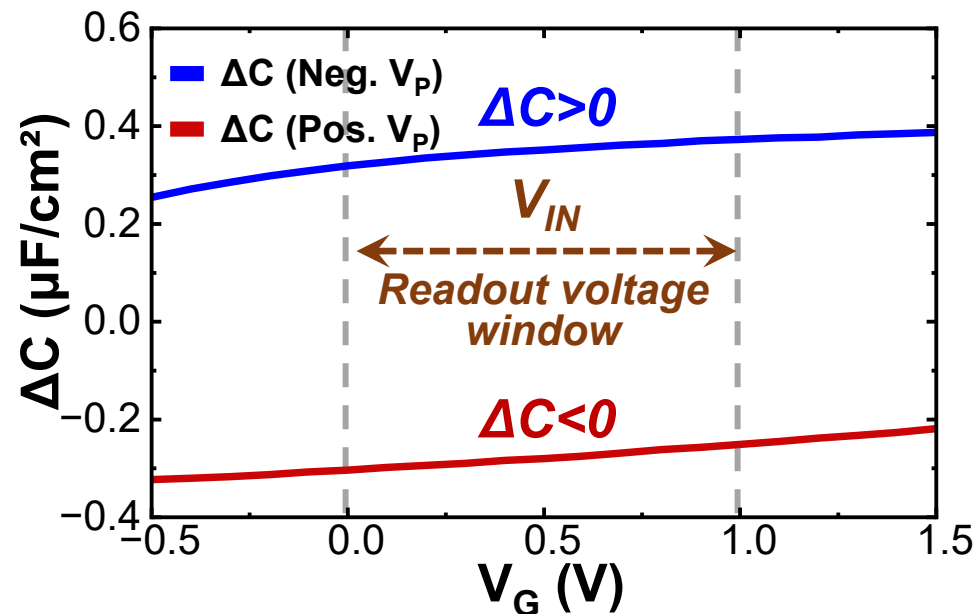
- Programmed C_{GP} and C_{GN} in FS-Cap
 - Positive program voltage ($+V_P$) $\rightarrow$ **Leftward** C-V curve shift
 - Negative program voltage ($-V_P$) $\rightarrow$ **Rightward** C-V curve shift



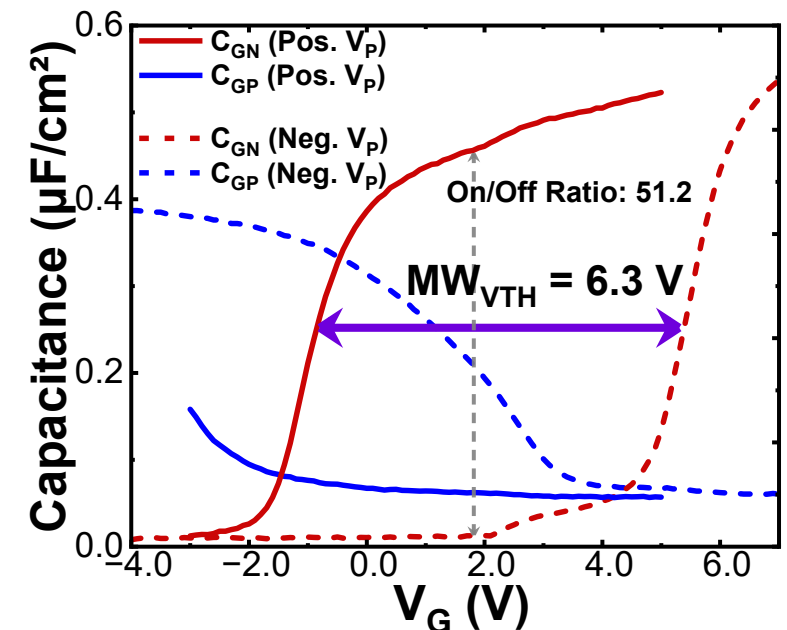
- ✓ Memory window of V_{TH} ($MW_{V_{TH}}$) is 6.3V
- ✓ Capacitance on/off ratio is 51.2

FS-Cap for ΔC storage (3/4)

- Spatial ΔC storage in FS-Cap
 - The dual-ended outputs of C_{GP} and C_{GN} exhibit complementarity
 - Enabling robust differential **ΔC memory window (ΔC_{MW})** through charge-steering

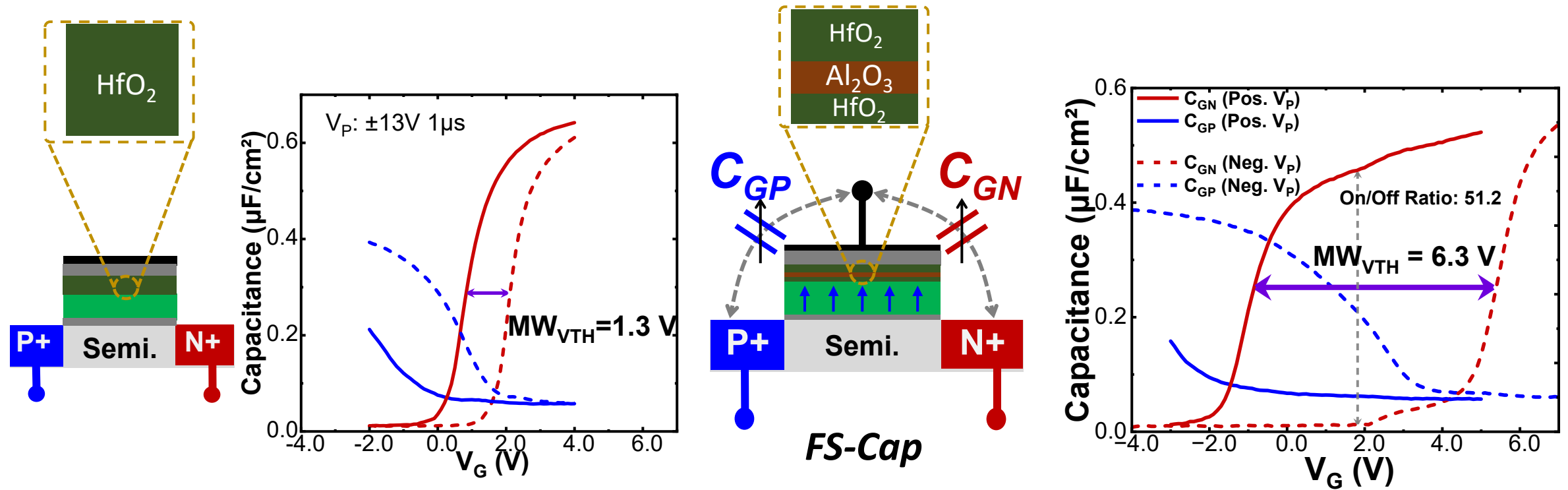


$$\Delta C = C_{GP} - C_{GN}$$



FS-Cap for ΔC storage (4/4)

- **Incorporation of gate trapping and barrier layer in FS-Cap**
 - Enhance trapping efficiency of gate-injected charge $\rightarrow$ improve $MW_{V_{TH}}$

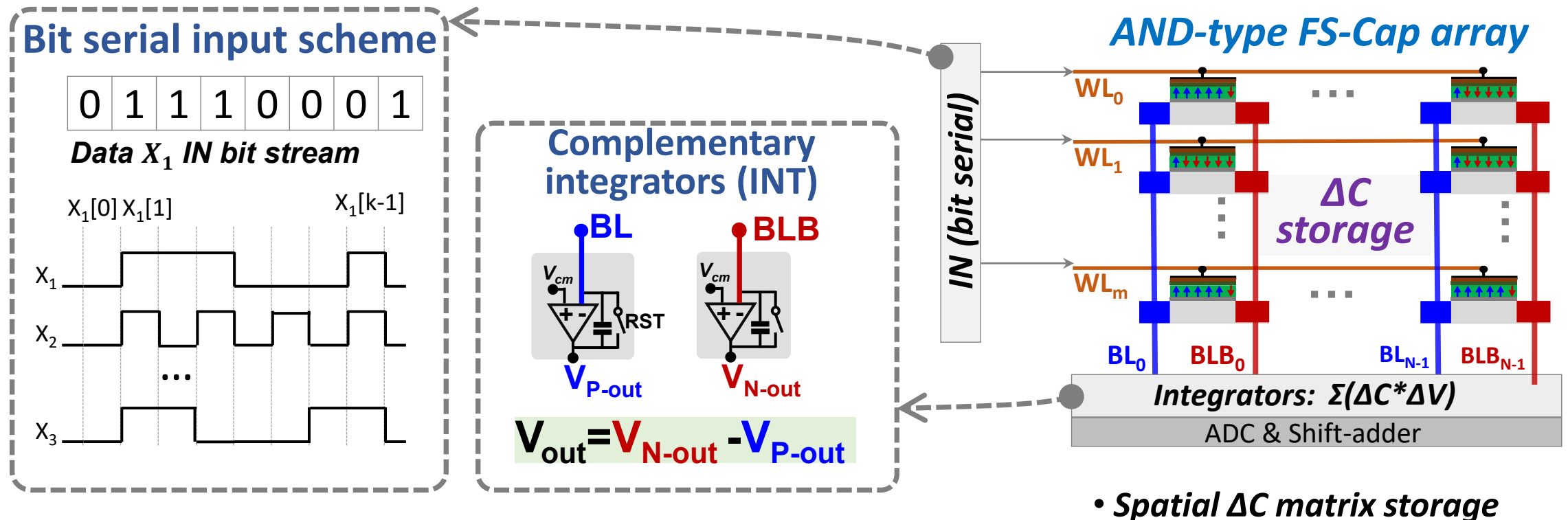


Outline

- FE charge-steering memcapacapacitor (FS-Cap) for Δ -CIM
- FS-Cap array based Δ -CIM behavior
 - FS-Cap array based Δ -CIM system
 - Δ -compute behavior for LLMs
- FS-Cap based Δ -CIM macro for LLMs
- Summary

FS-Cap array based CIM system

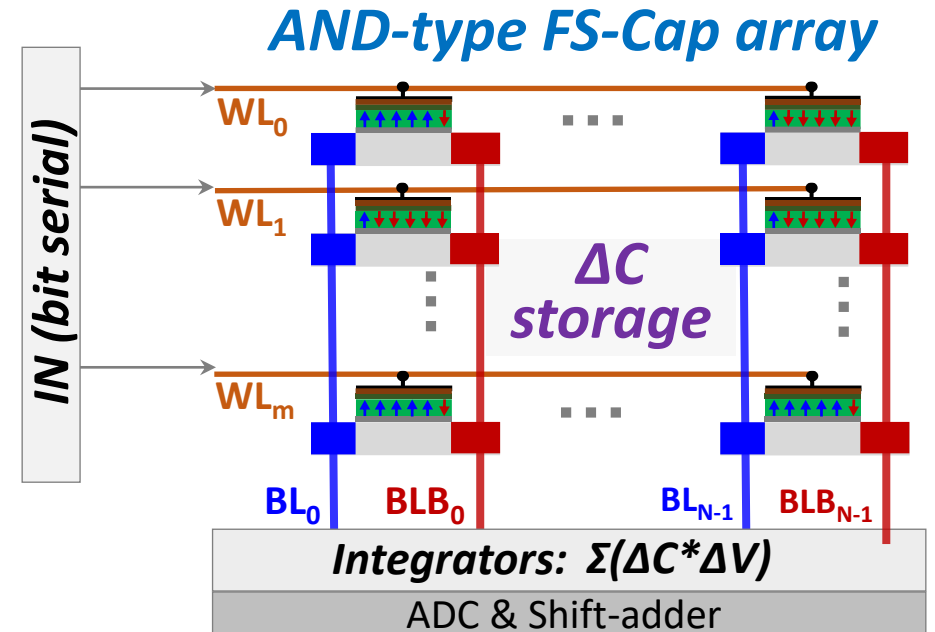
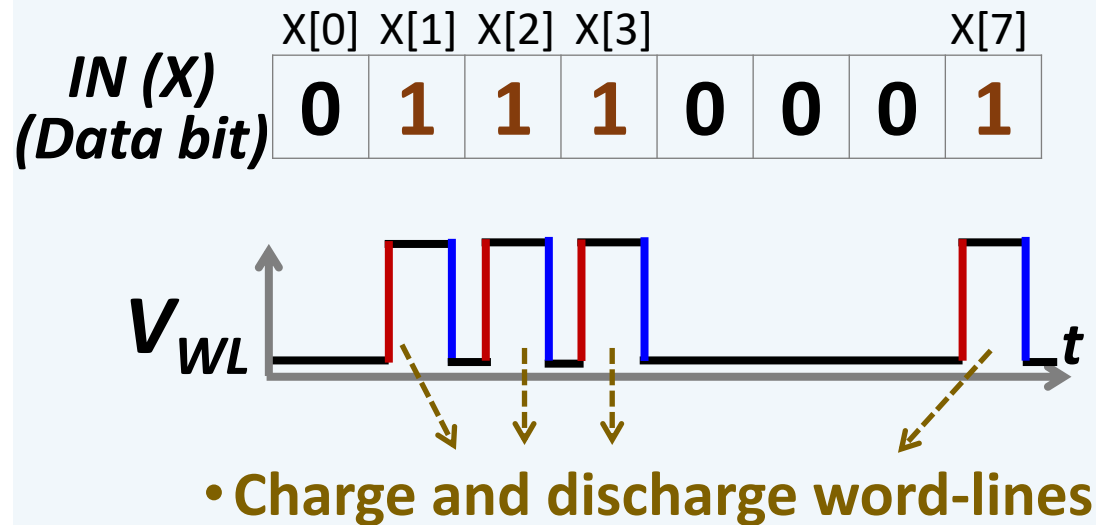
- FS-Cap array enables **charge-steering based differential computing**
 - Achieve inherent common-mode noise rejection (CMNR)



ΔIN encoding for FS-Cap-based Δ -CIM (1/3)

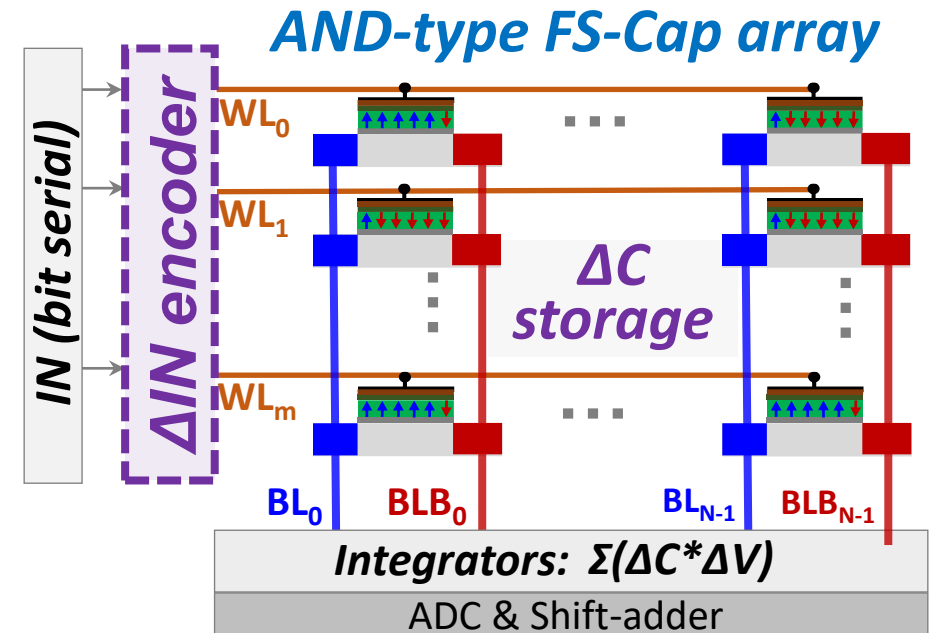
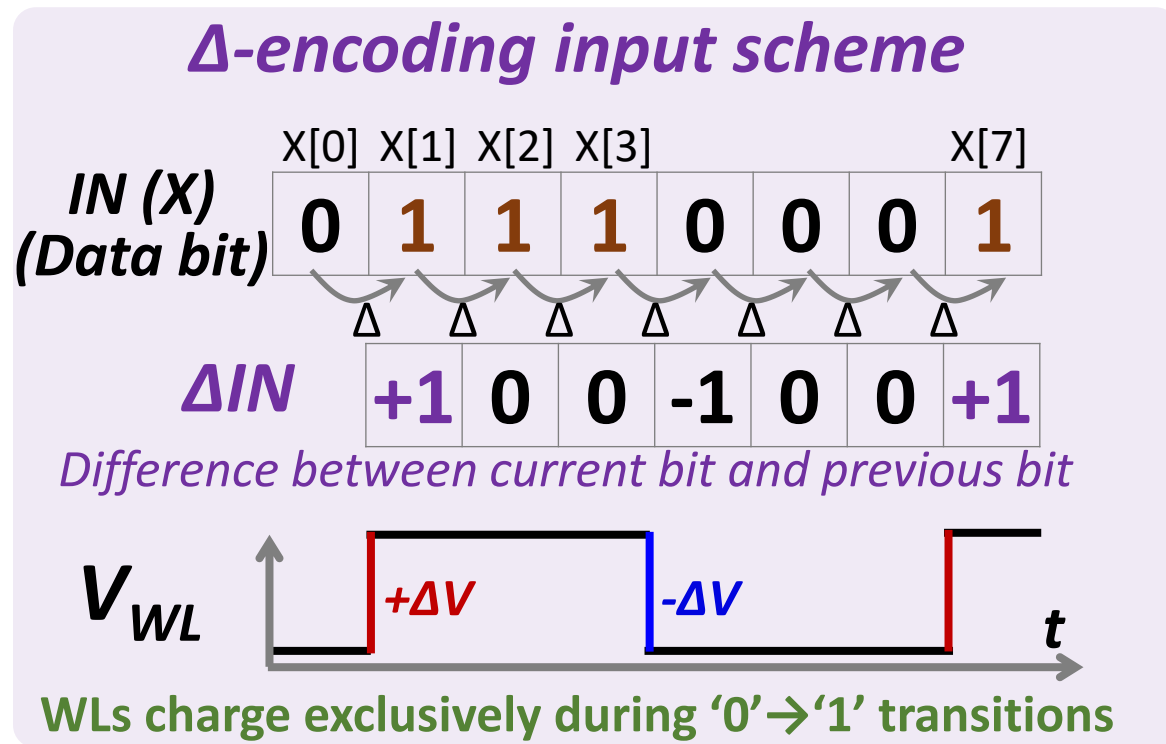
- The conventional bit-by-bit input scheme
 - Per-cycle WL charging even for identical bit $\rightarrow$ Redundant dynamic power

Conventional input scheme



Δ IN encoding for FS-Cap-based Δ -CIM (2/3)

- **This work:** temporal delta-encoding for input: $\Delta IN = IN_{[t]} - IN_{[t-1]}$
 - Modulate input as bit-transition signals $\rightarrow$ Event-driven computation



ΔIN encoding for FS-Cap-based Δ -CIM (3/3)

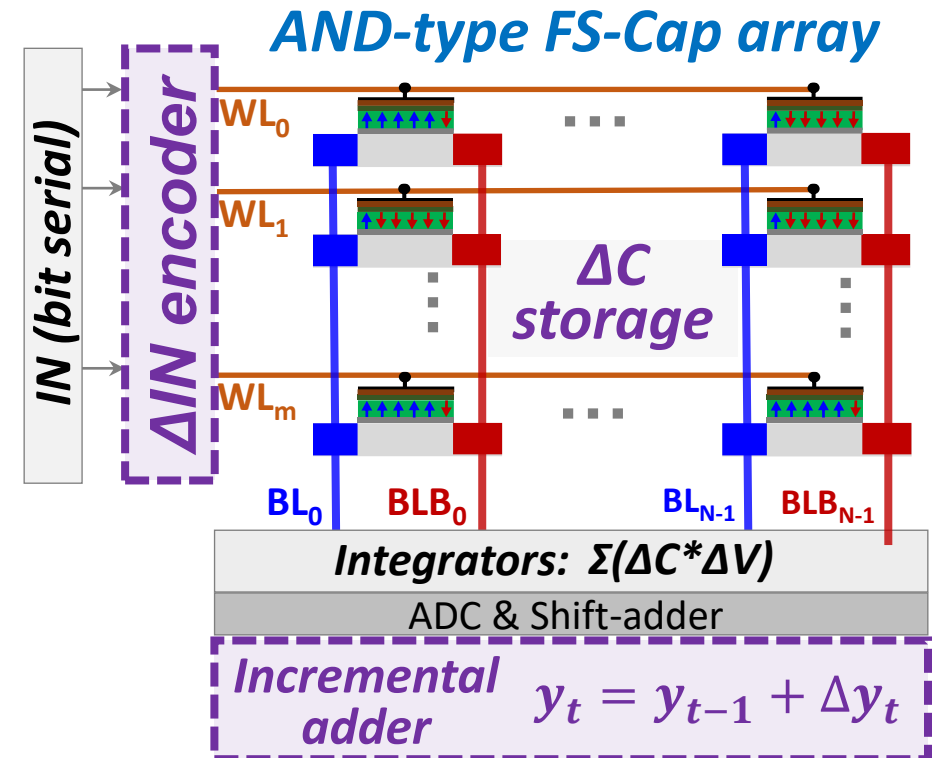
- Temporal ΔIN based MAC and searching computation
 - Original compute results is obtained through incremental adder

$IN[t-1]$	$IN[t]$	ΔIN	WL	ΔQ_{out}
0	0	0	V_{WL} stable	0
0	1	+1	$+\Delta V$	$+\Delta C \cdot \Delta V$
1	0	-1	$-\Delta V$	$-\Delta C \cdot \Delta V$
1	1	0	V_{WL} stable	0

ΔIN generates ΔQ_{out} @ per transition event

$$\Delta IN = IN[t] - IN[t - 1]$$

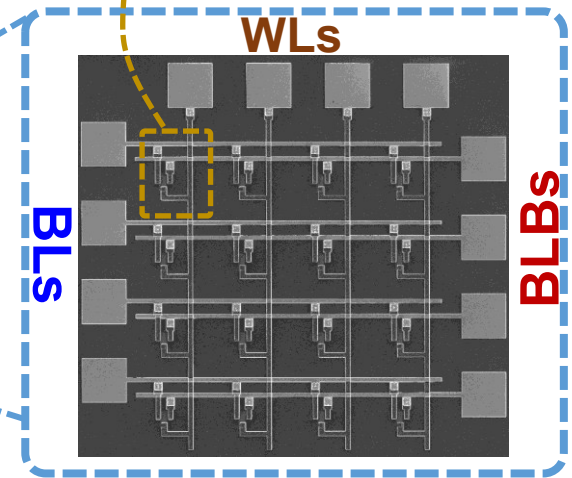
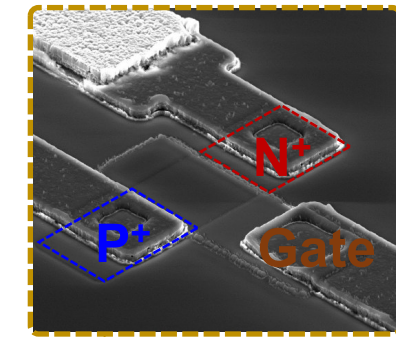
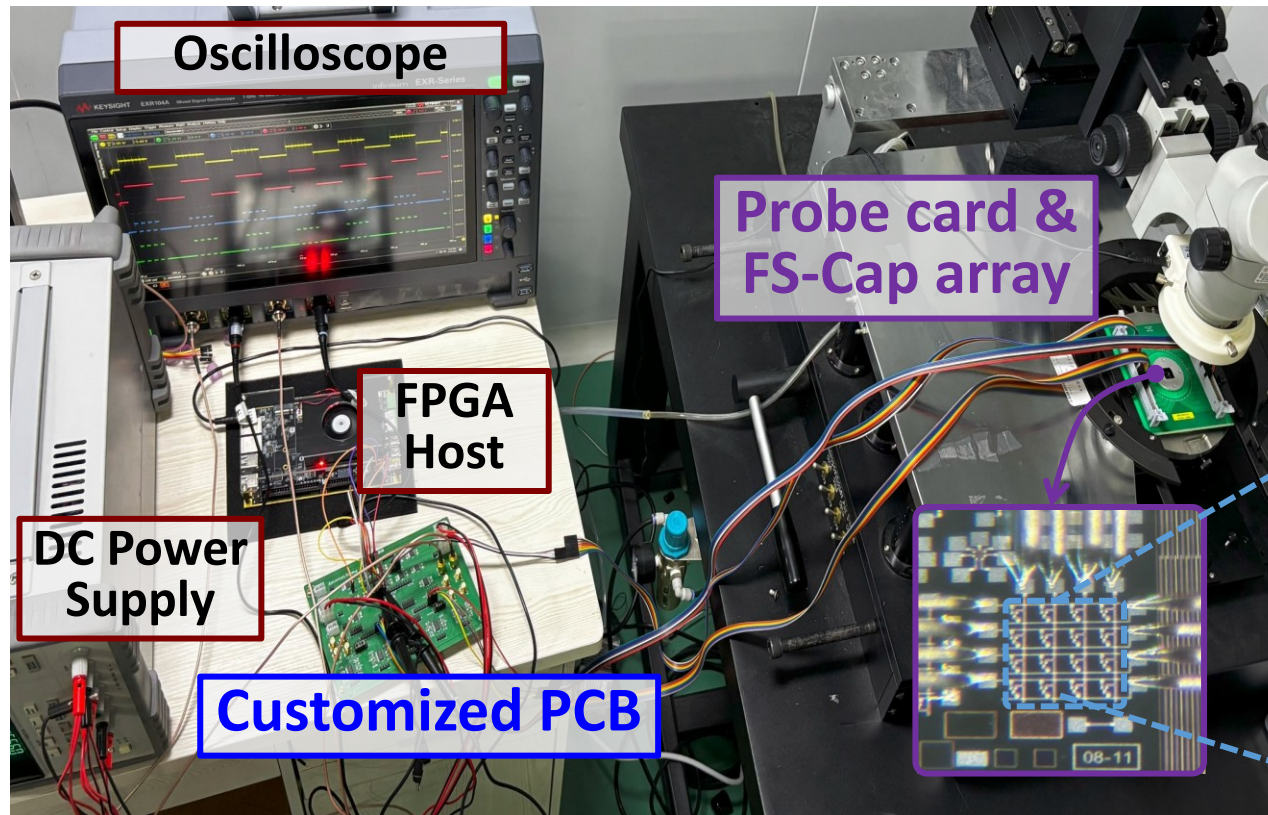
$$Q_{out}[t] = Q_{out}[t - 1] + \alpha \cdot \Delta Q_{out}[t]$$



Outline

- FE charge-steering memcapacapacitor (FS-Cap) for Δ -CIM
- FS-Cap array based Δ -CIM behavior
 - FS-Cap array based Δ -CIM system
 - Δ -compute behavior for LLMs
- FS-Cap based Δ -CIM macro for LLMs
- Summary

Experimental setup of FS-Cap array based Δ -CIM system

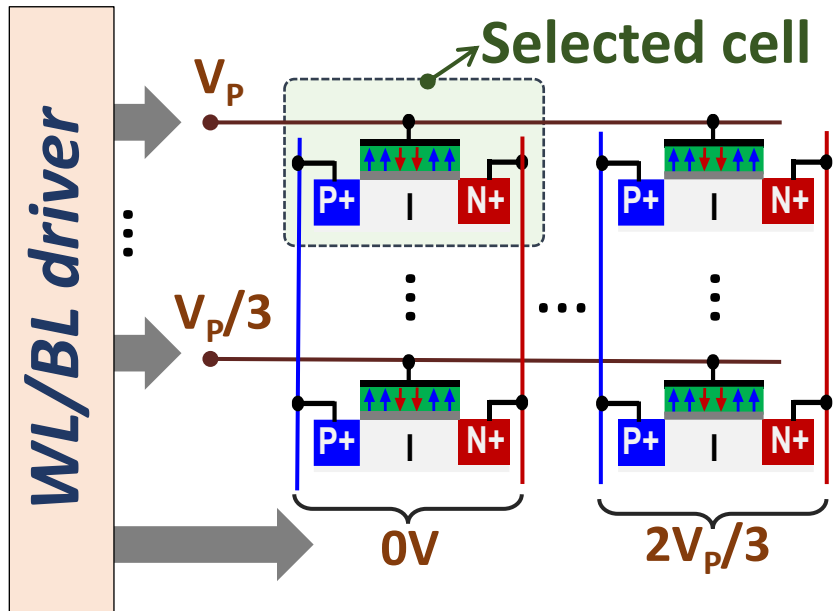


(4WLs, 4BLs, 4BLBs)

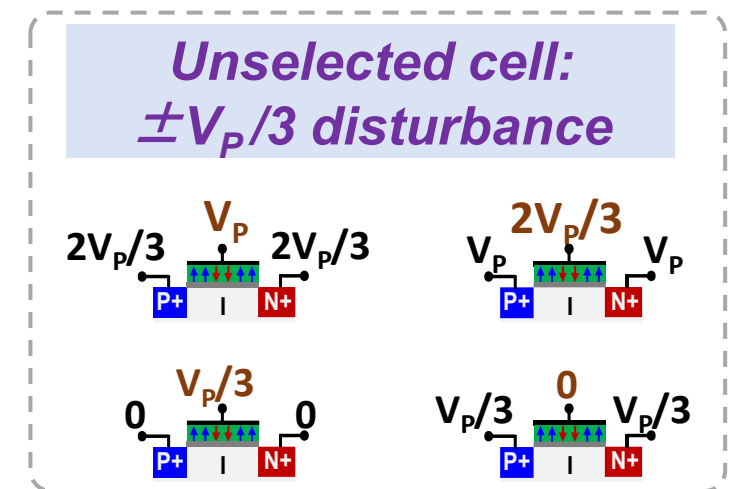
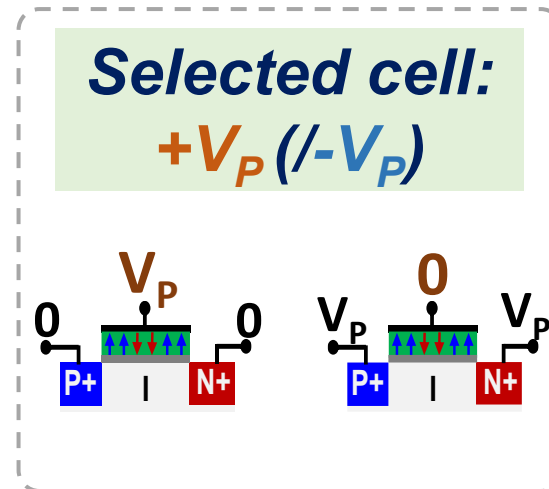
ΔC programming in AND-type FS-Cap array (1/3)

- The $V_p/3$ program scheme for cell-selective ΔC configuration

AND-type FS-Cap array

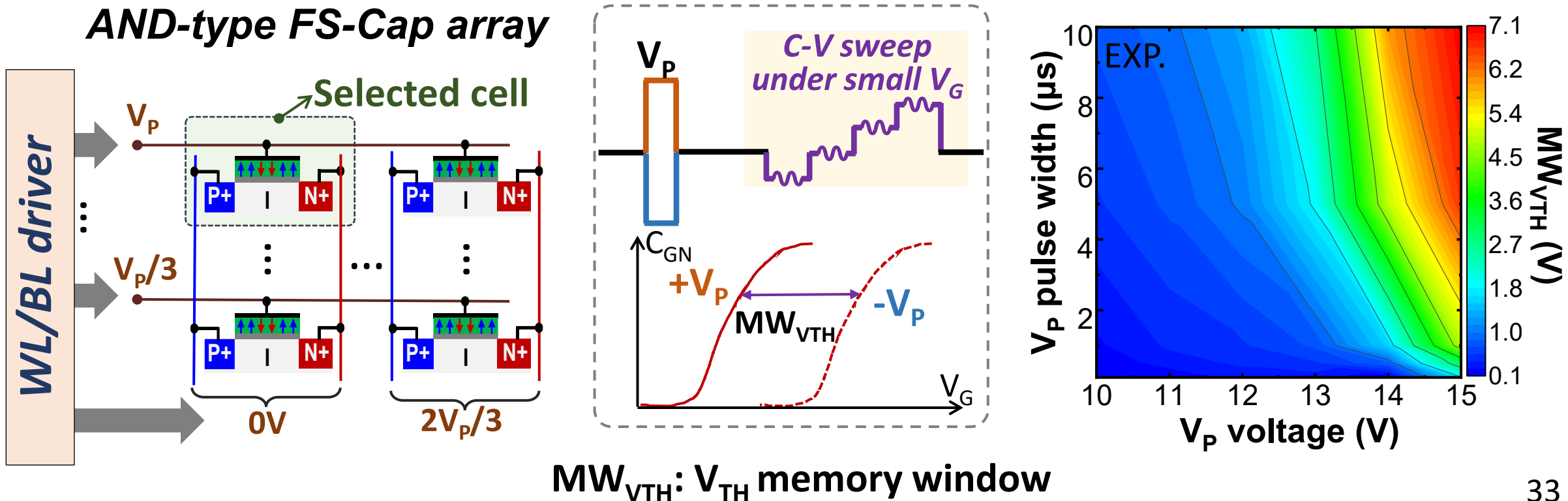


- $V_p \rightarrow$ selected row/column
- $V_p/3 \rightarrow$ unselected rows/columns
- $2V_p/3 \rightarrow$ unselected columns/rows



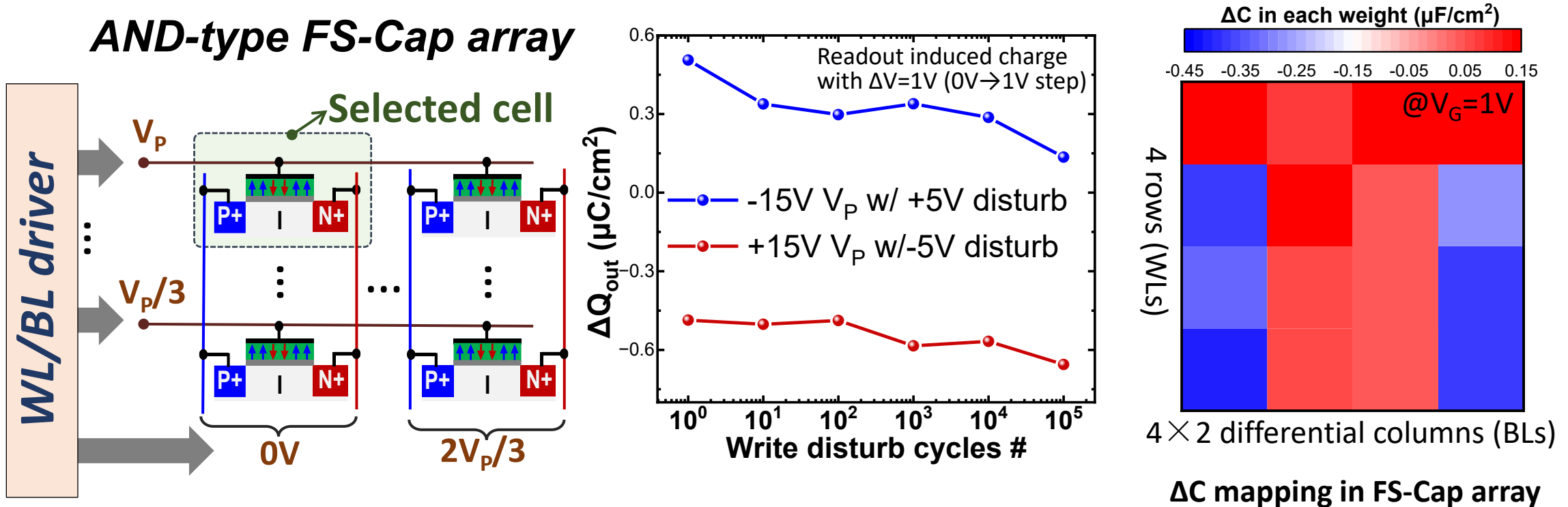
ΔC programming in AND-type FS-Cap array (2/3)

- $MW_{V_{TH}}$ of FS-Cap depends on voltage and width of V_p pulse
 - Effectively prevent write disturbance under $V_p/3$



ΔC programming in AND-type FS-Cap array (3/3)

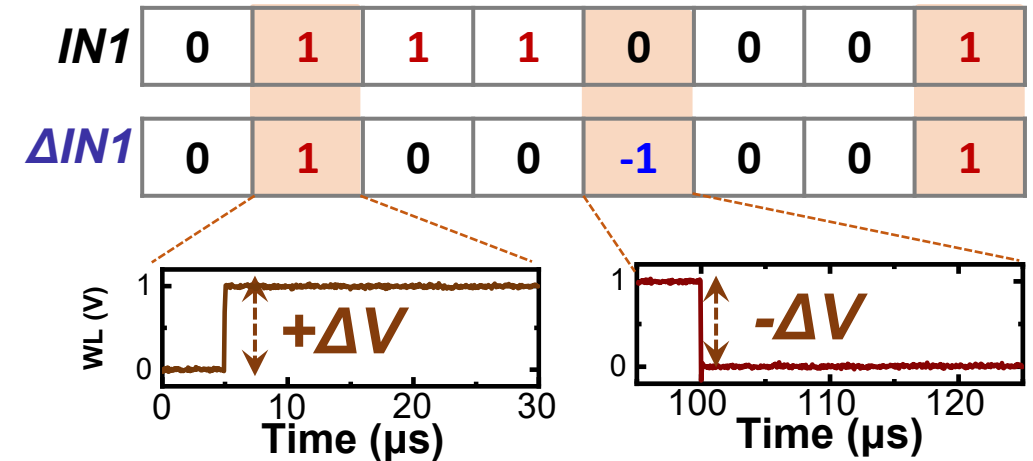
- ΔC array programming
 - FS-Cap maintain ΔQ_{out} due to robust ΔC_{MW} under $\pm V_p/3$ disturbances



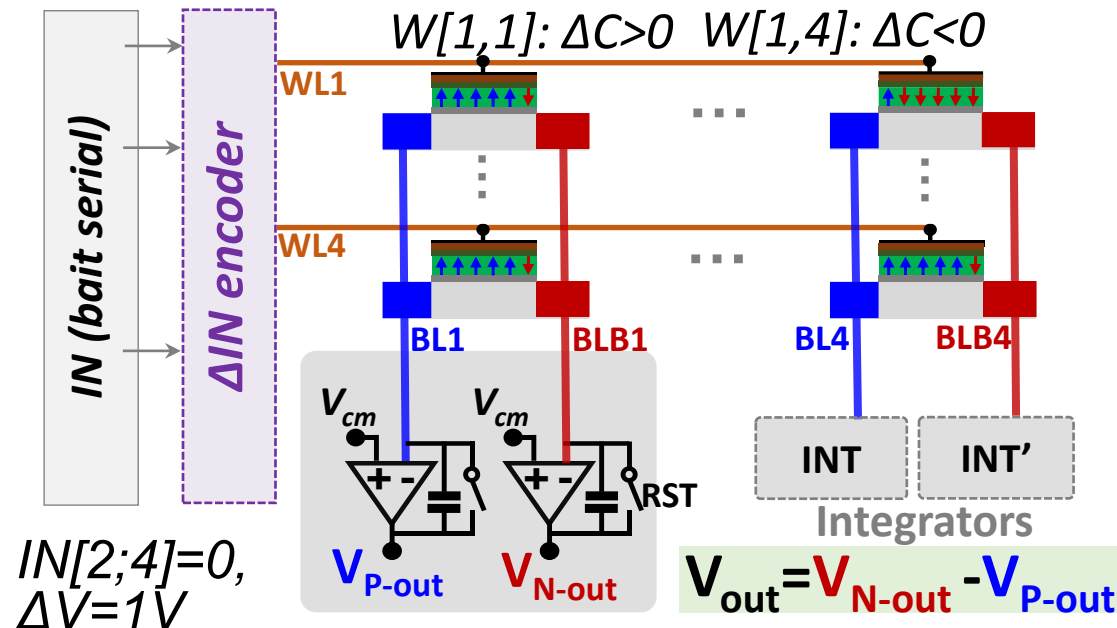
Δ -MAC for dot-product attention (1/5)

— Δ IN encoding

- Input vectors undergo Δ IN encoding
 - WL charging ($+\Delta V$): ' $0 \rightarrow 1$ ' transitions
 - WL discharging ($-\Delta V$): ' $1 \rightarrow 0$ ' transitions
 - $\Delta IN = '0' \rightarrow$ No charging/discharging



✓ *Transition-triggered WL signals*

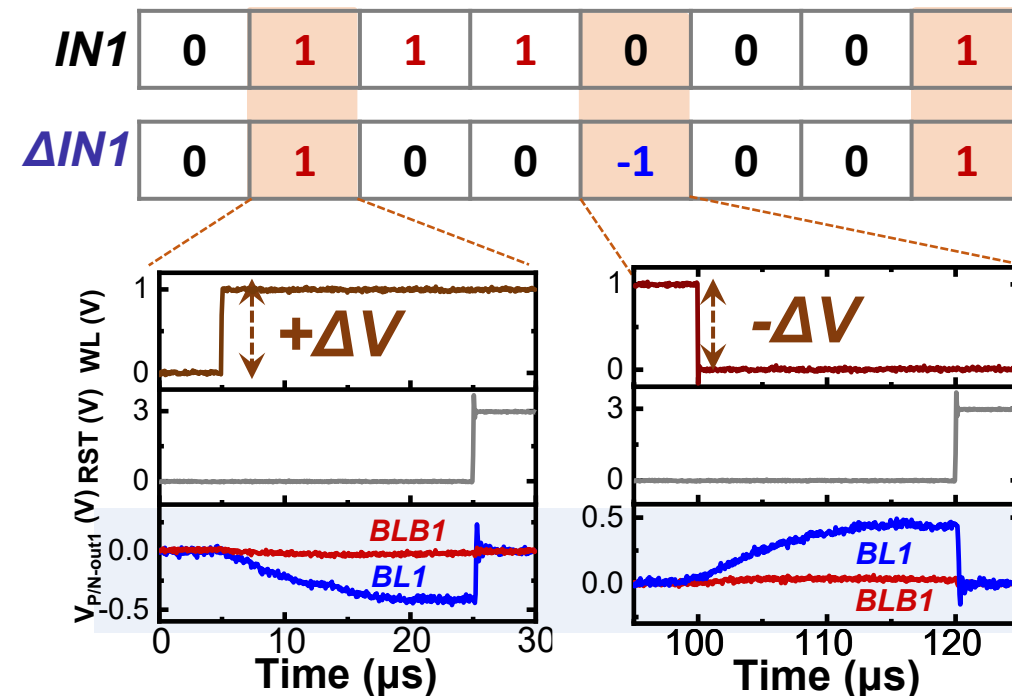
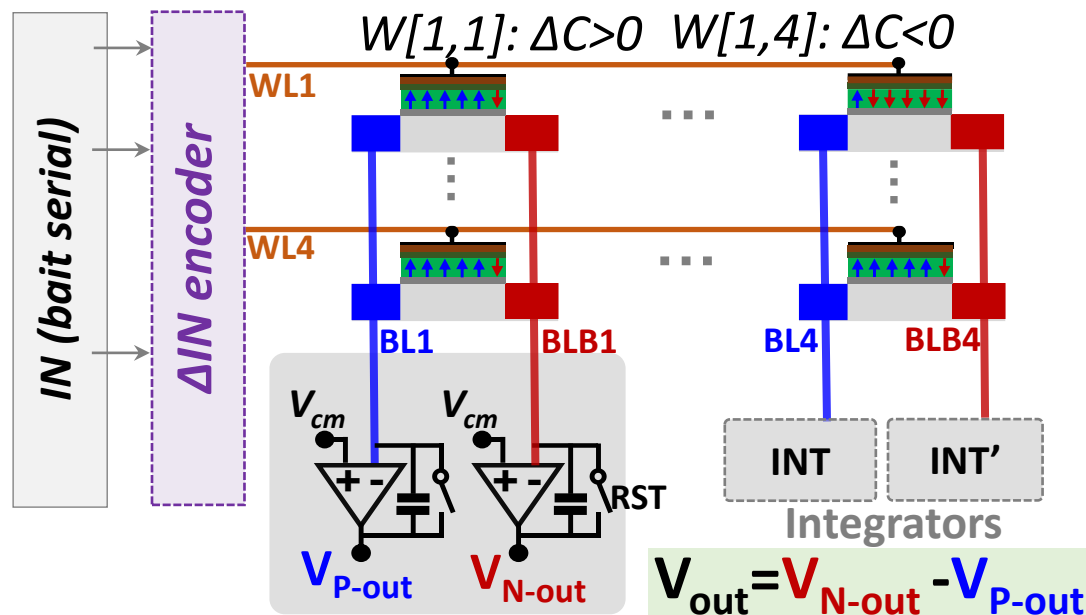


Δ -MAC for dot-product attention (2/5)

— Δ -Multiplication

- For $\Delta C > 0$

- BL : high $C_{GP} \rightarrow$ high $|V_{P-out}|$
- BLB: low $C_{GN} \rightarrow$ low $|V_{N-out}|$

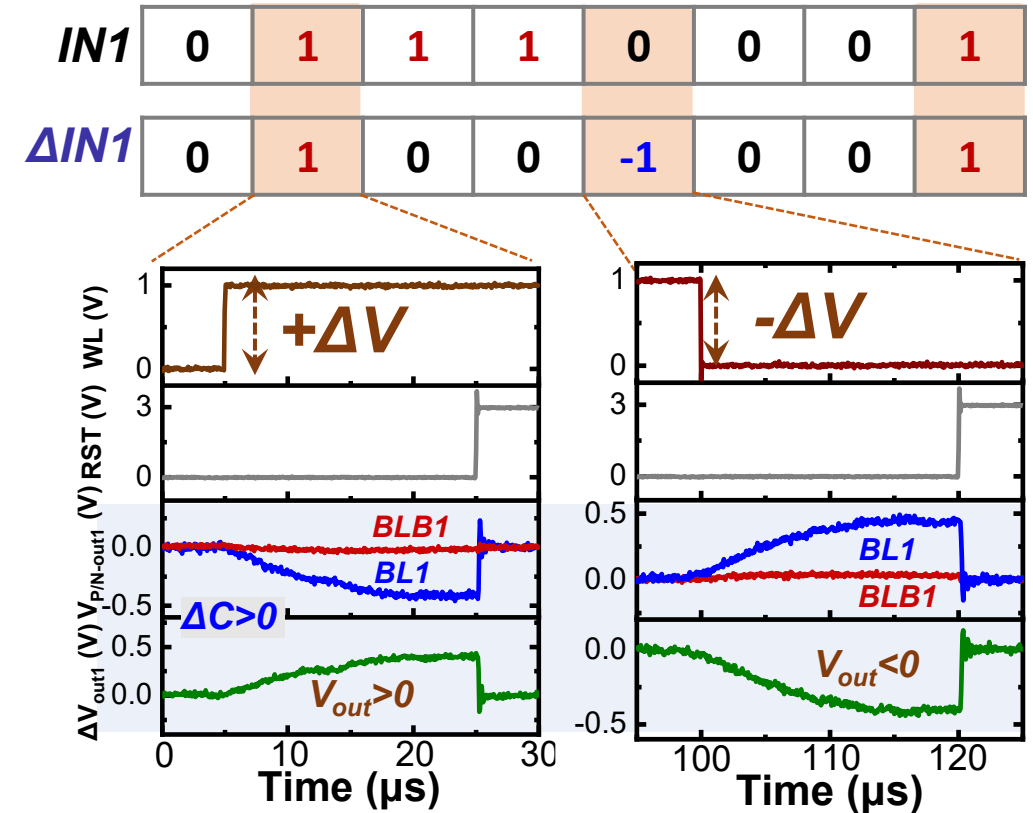
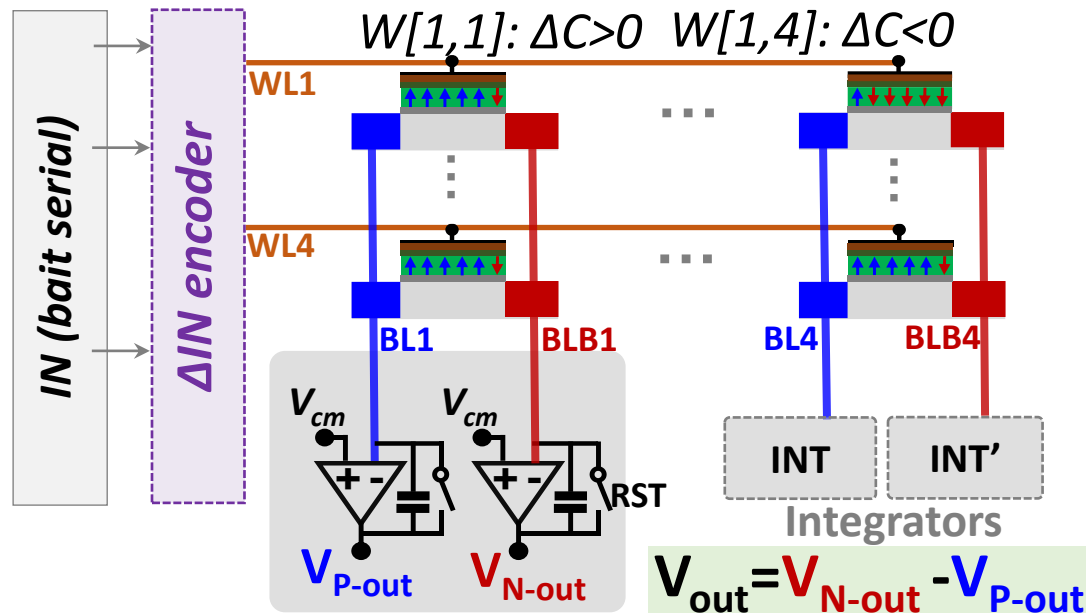


- The induced charge is steered to complementary BL/BLB according to the stored ΔC states.
- The integrators generate output voltage (V_{out}), representing the Δ -MAC result in this cycle

Δ -MAC for dot-product attention (3/5)

— Δ -Multiplication

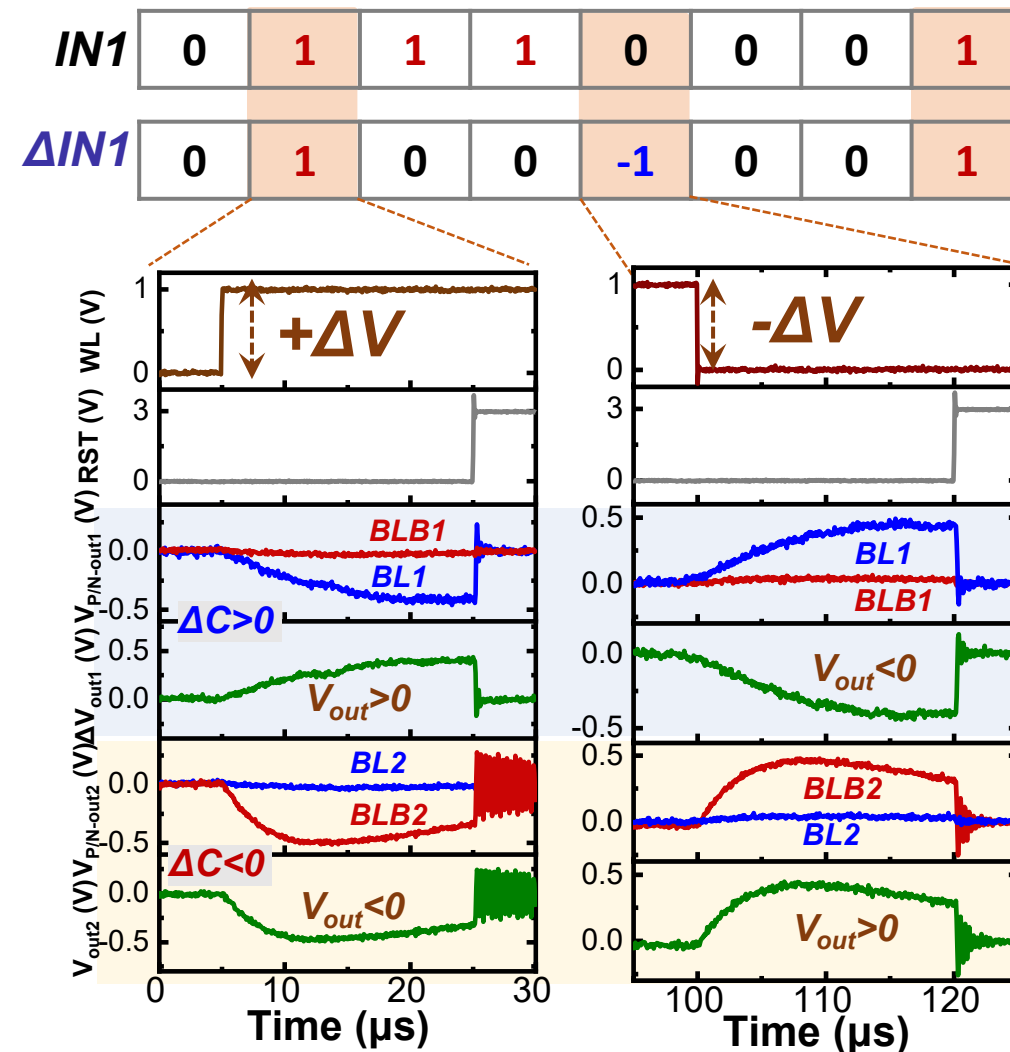
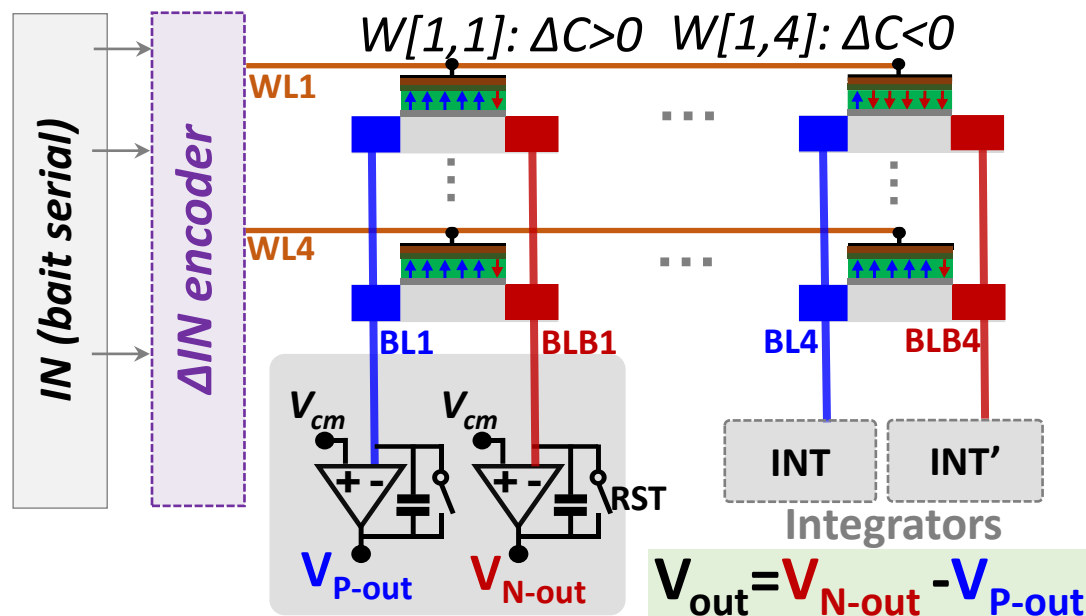
- For $\Delta C > 0$
 - BL : high $C_{GP} \rightarrow$ high $|V_{P-out}|$
 - BLB: low $C_{GN} \rightarrow$ low $|V_{N-out}|$
 - V_{out} polarity depends on ΔIN



Δ -MAC for dot-product attention (4/5)

— Δ -Multiplication

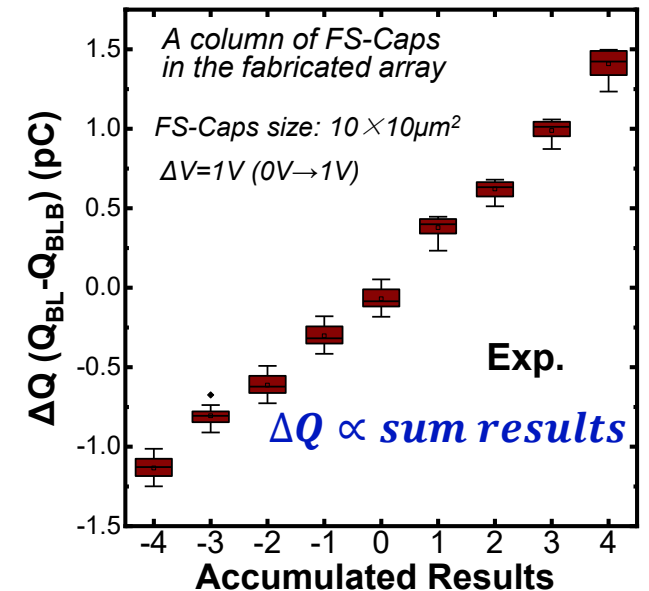
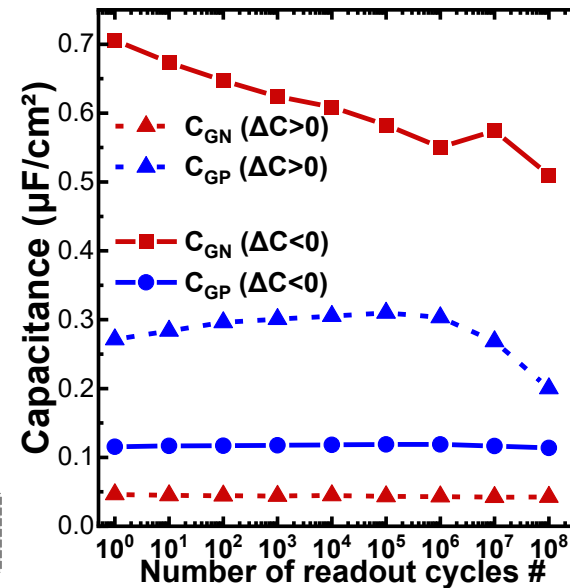
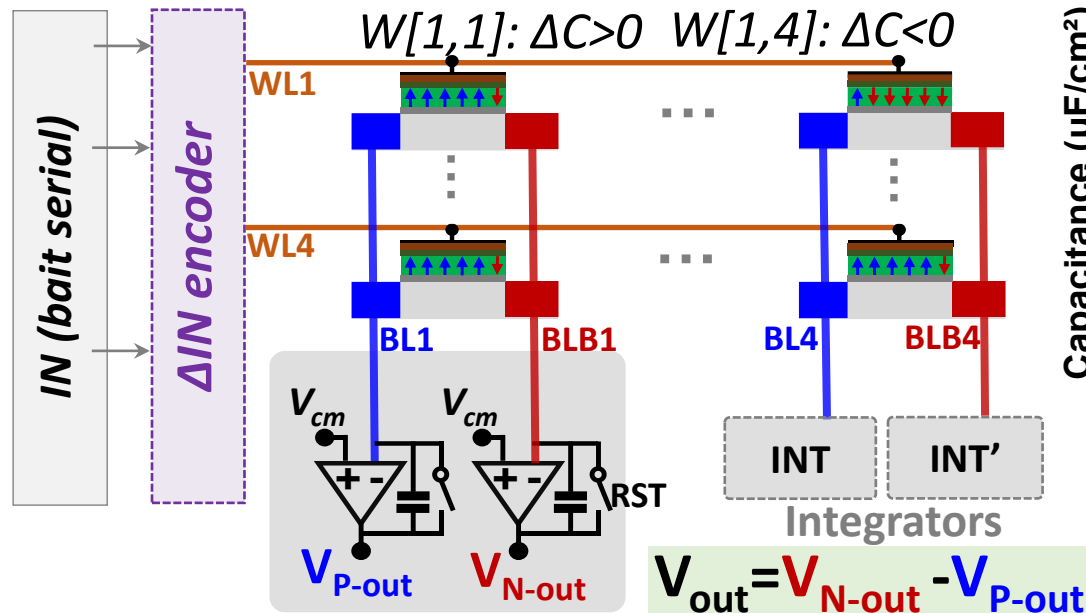
- For $\Delta C > 0$
- For $\Delta C < 0$
 - Low $|V_{P-out}|$; high $|V_{N-out}|$
 - V_{out} polarity is opposite to that of $\Delta C > 0$



Δ -MAC for dot-product attention (5/5)

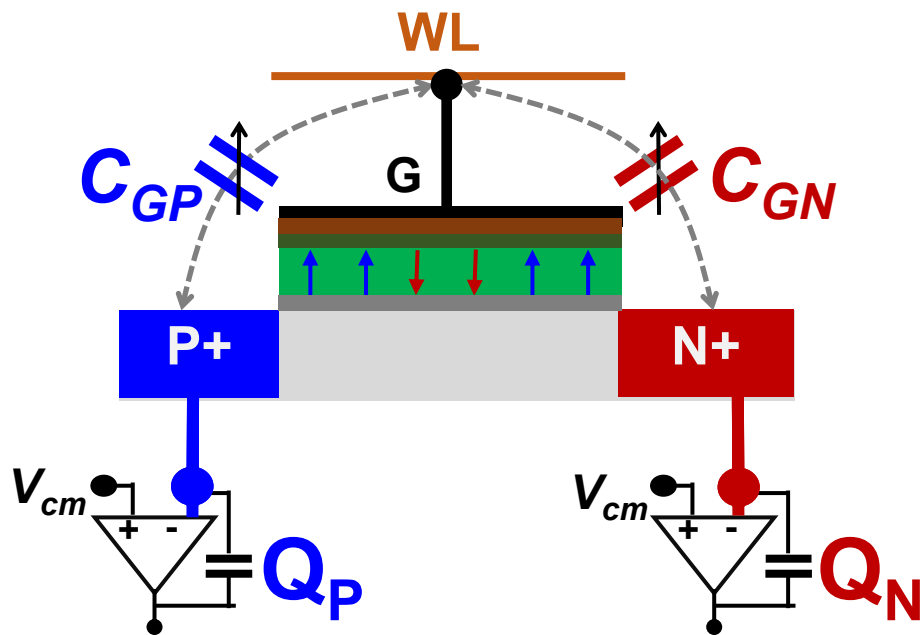
— Δ -Accumulation

- The ΔC states remains stable during cycling computing ($>10^8$)
- The cumulative charge exhibits a linear relationship with the MAC results



Δ -CAM for K - V cache searching in LLMs (1/2)

- CAM mode: Measure Q-K Hamming distance for relevant K - V cache search
- Query ($\pm V_{IN}$) $\rightarrow$ Q_{out} polarity indicate match/mismatch



Searching mode

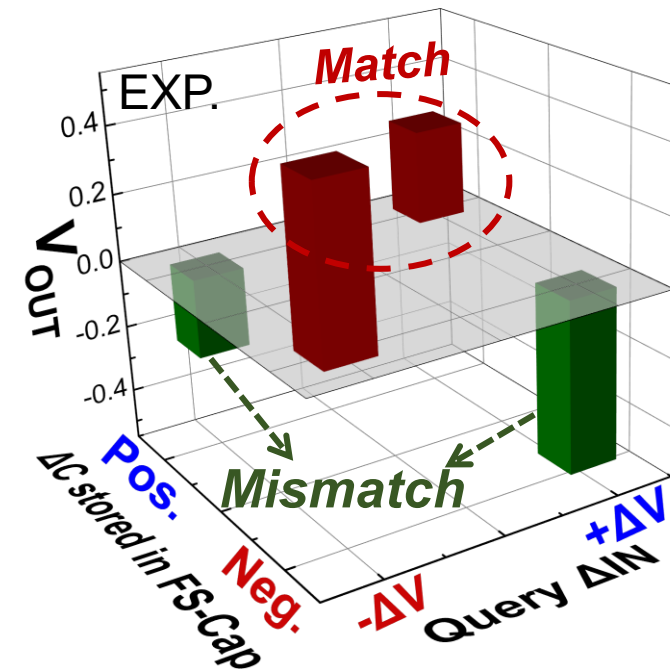
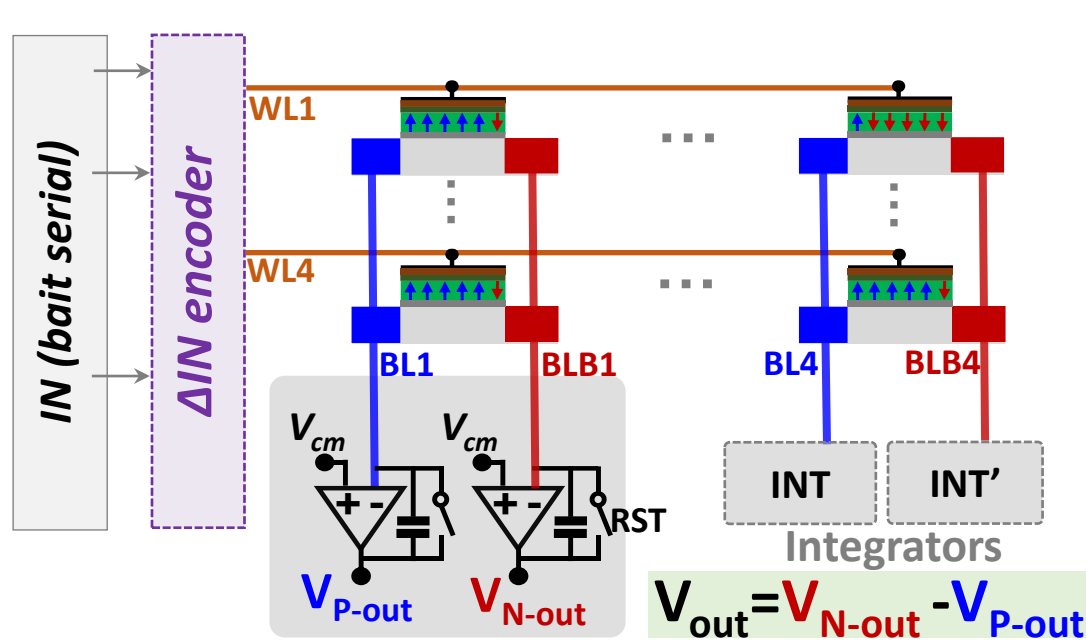
Entry	$IN_{Query} (V_{WL})$	Q_{out}
"0" ($\Delta C < 0$)	"0" ($-\Delta V$)	+
	"1" ($+\Delta V$)	-
"1" ($\Delta C > 0$)	"0" ($-\Delta V$)	-
	"1" ($+\Delta V$)	+

XNOR of content-addressable memory (CAM)

$$Q_{out} = Q_P - Q_N$$

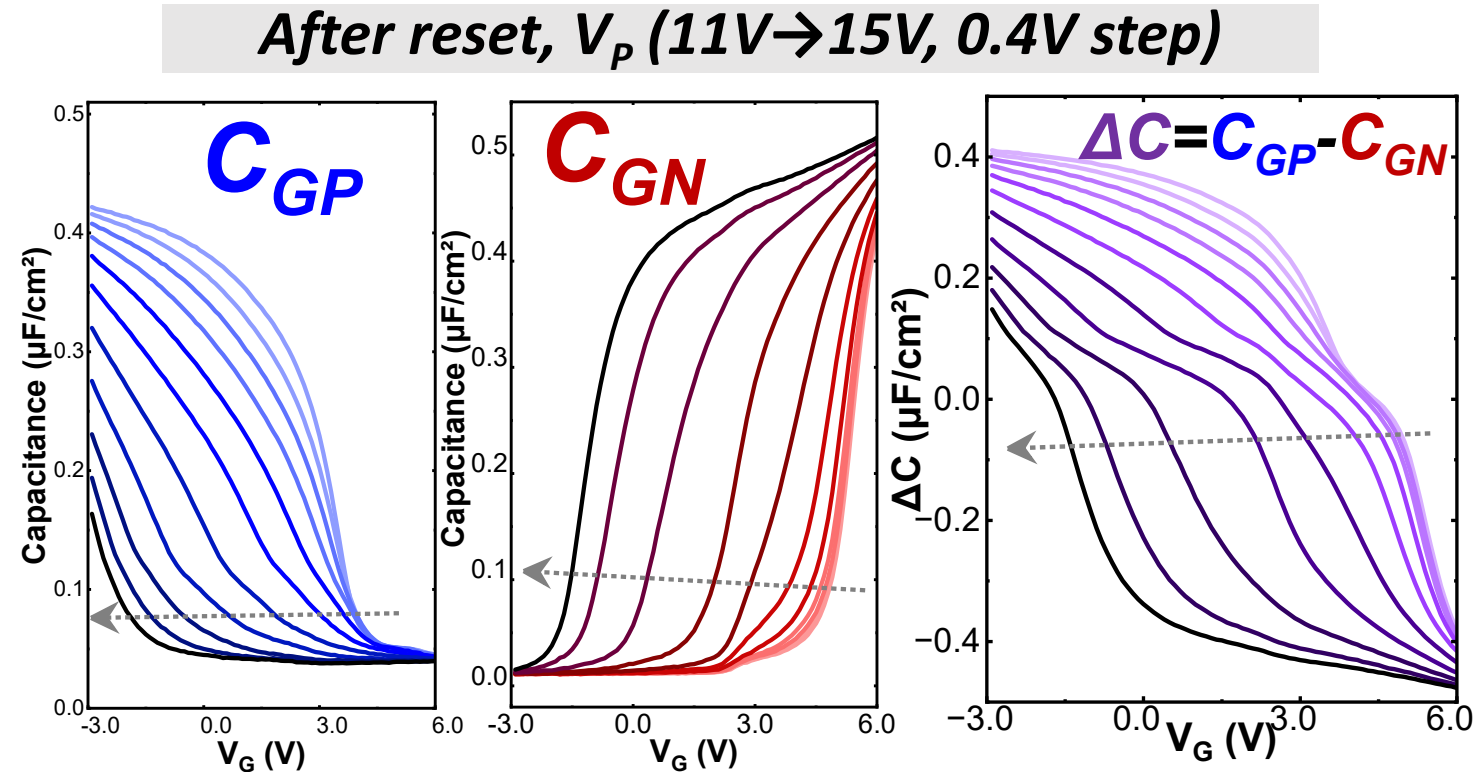
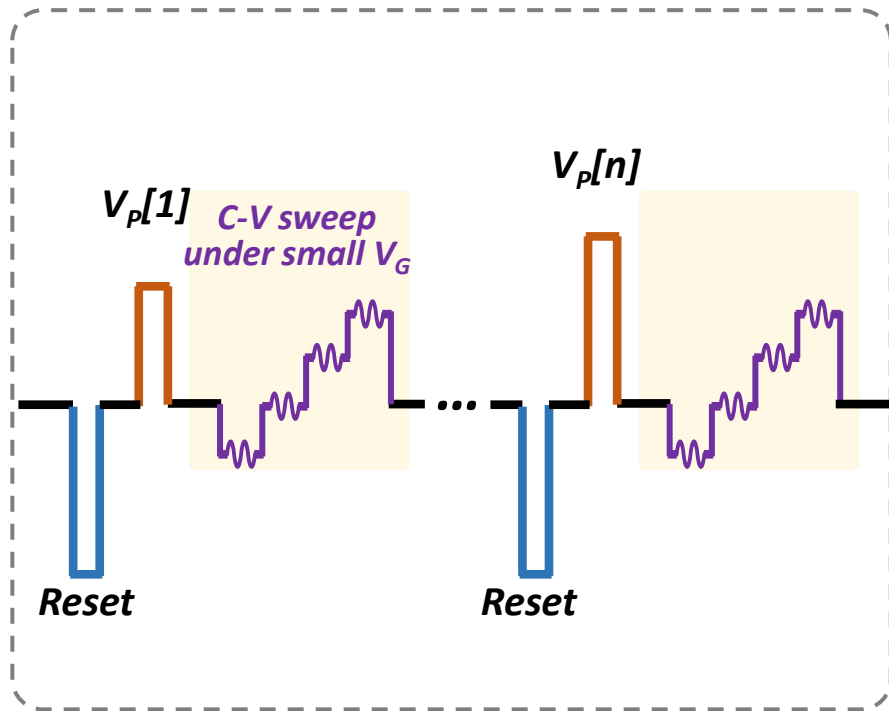
Δ -CAM for K - V cache searching in LLMs (2/2)

- **Charge-steering mechanism** enables CAM-based searching
 - Inherently implement bitwise XNOR in each FS-Cap



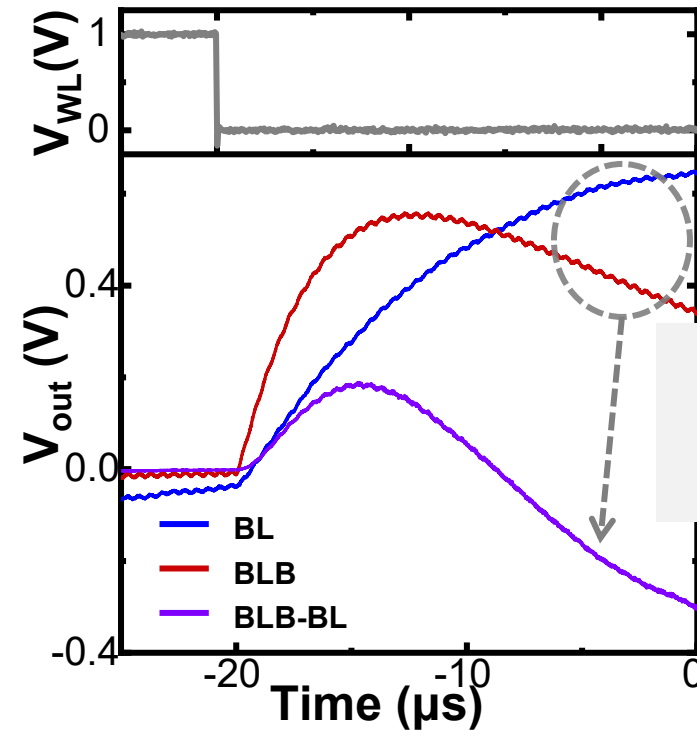
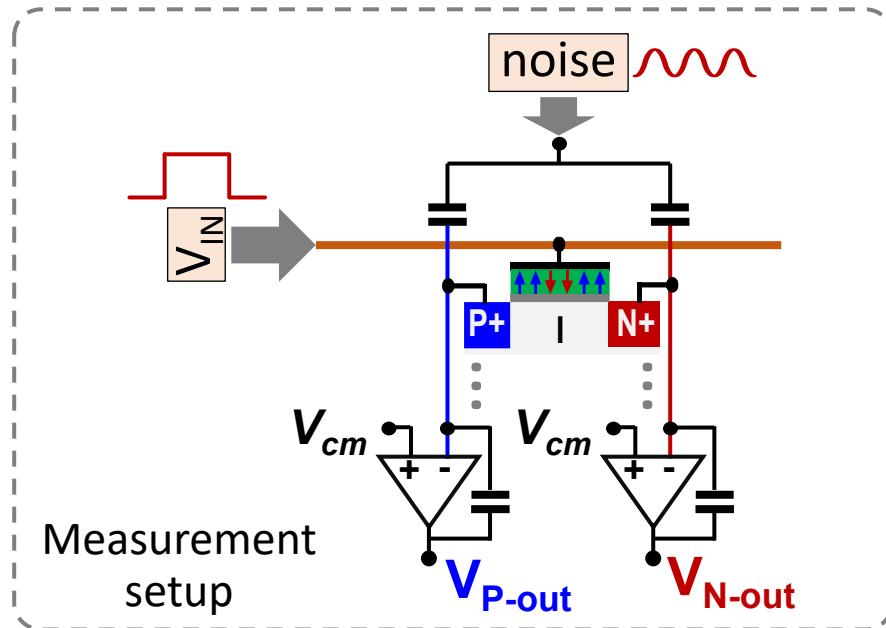
Experimental analysis of multi-level ΔC

- FS-Cap exhibits multi-level ΔC storage



Experimental analysis of noise suppression

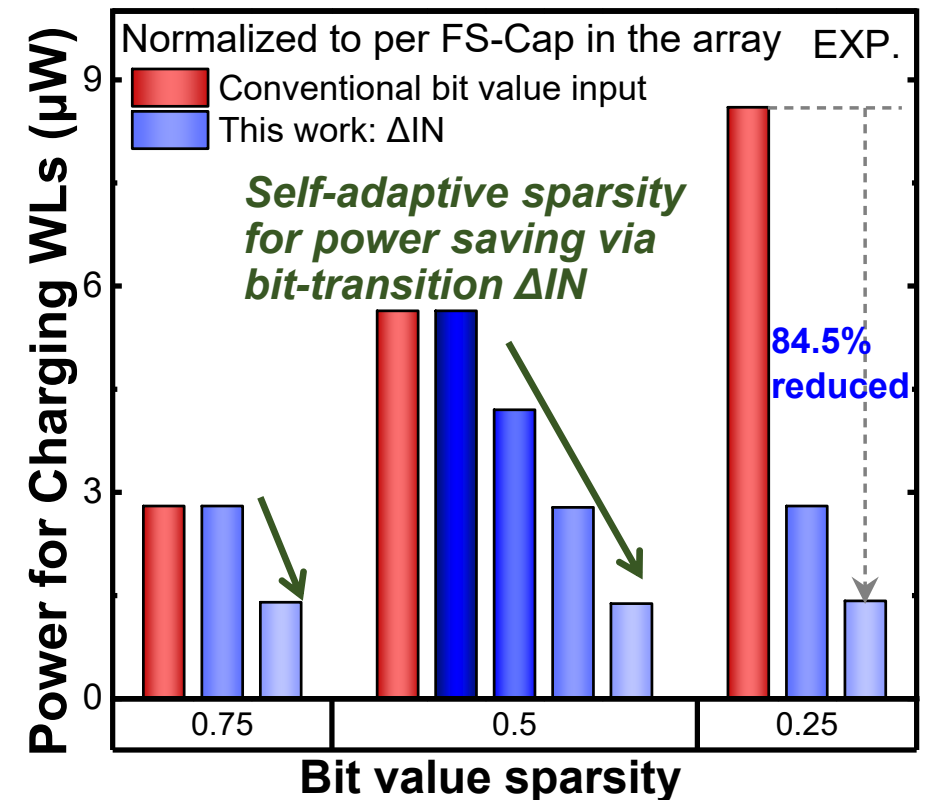
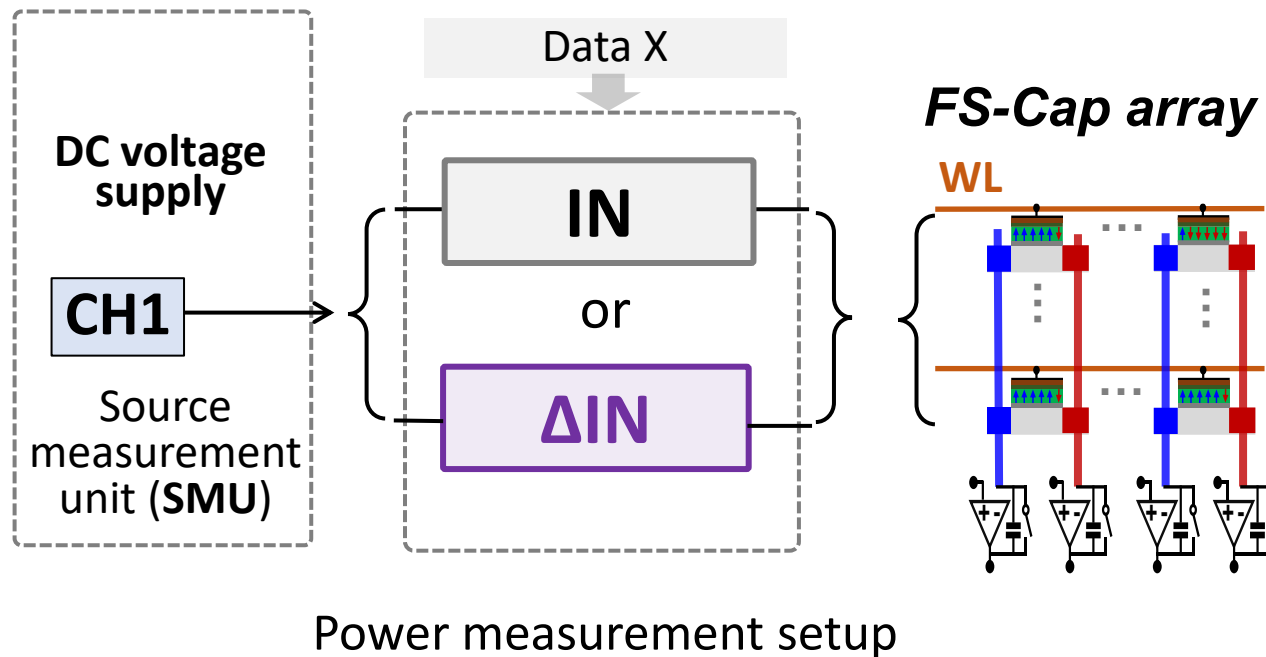
- The ΔC -based differential design improve SNR versus single-ended C-CIM
 - Enables higher parallel computing throughput for LLMs



Reject common-mode noise in BL/BLB lines

Experimental analysis of power efficiency

- Δ -CIM significantly improves energy efficiency
 - Save computing energy due to the bit-transition driven sparsity adaptation

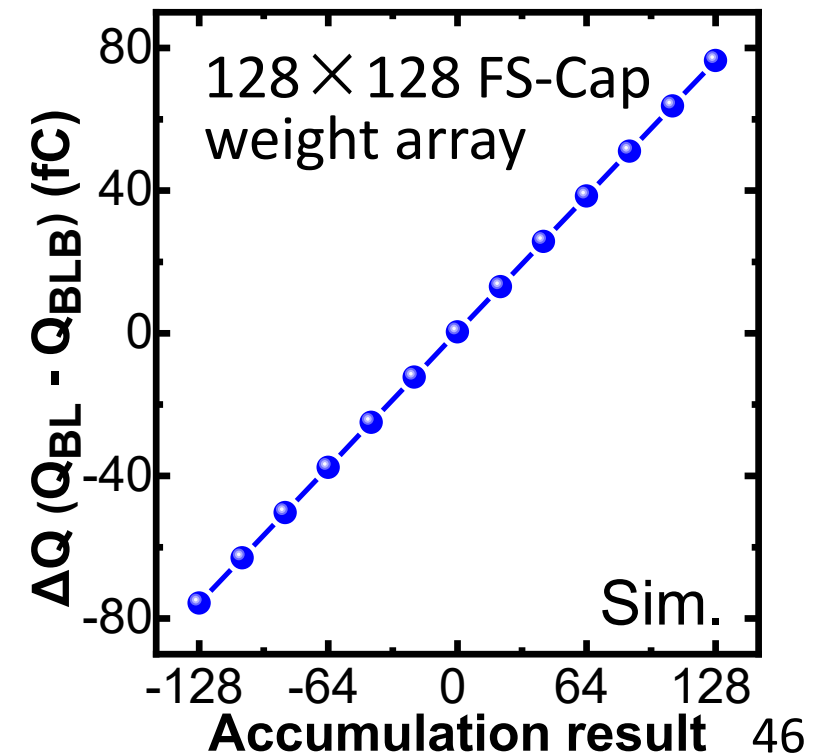
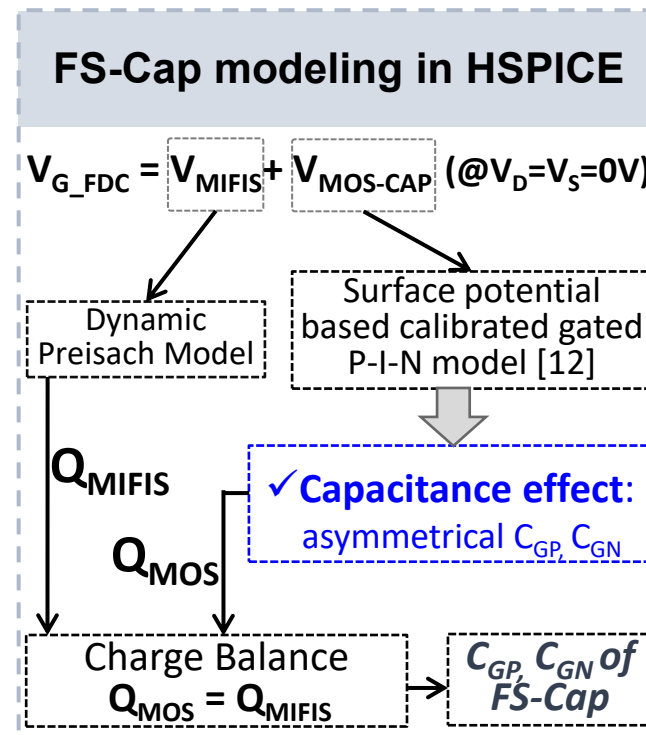
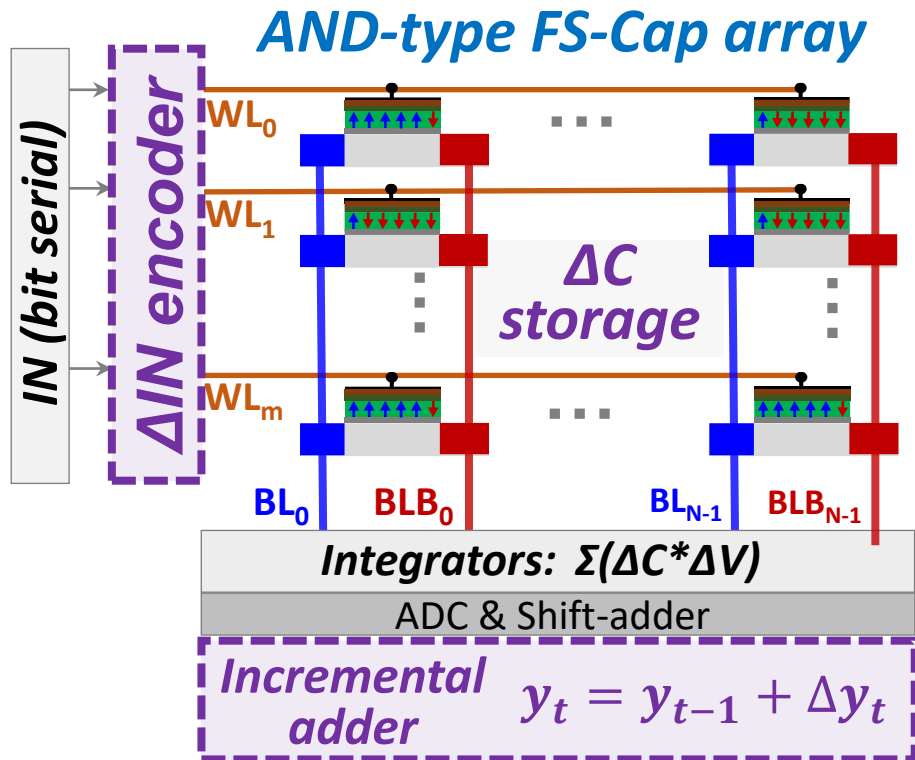


Outline

- FE charge-steering memcapacitor (FS-Cap) for Δ -CIM
- FS-Cap array based Δ -CIM behavior
- FS-Cap based Δ -CIM macro for LLMs
- Summary

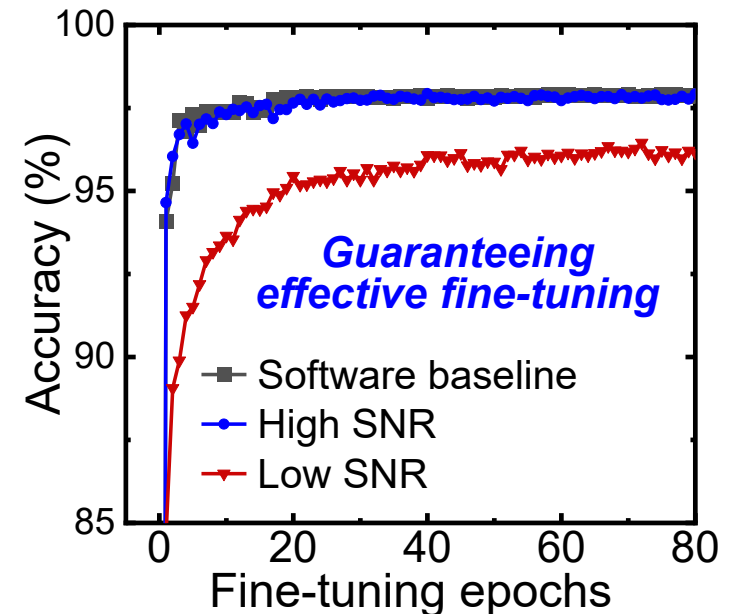
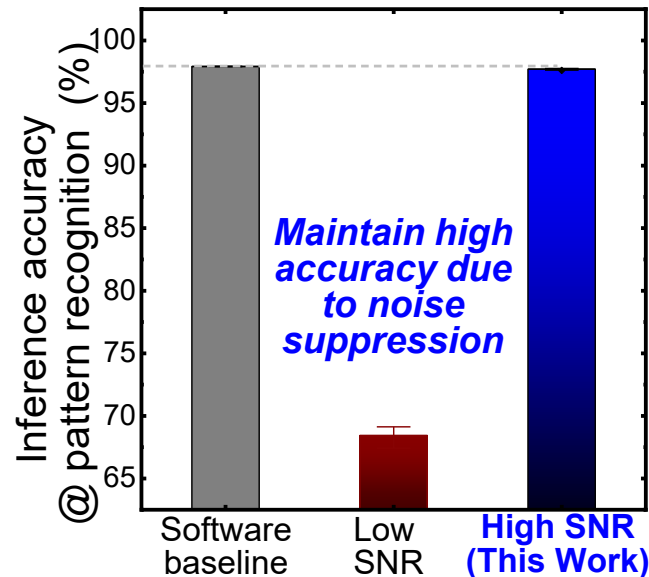
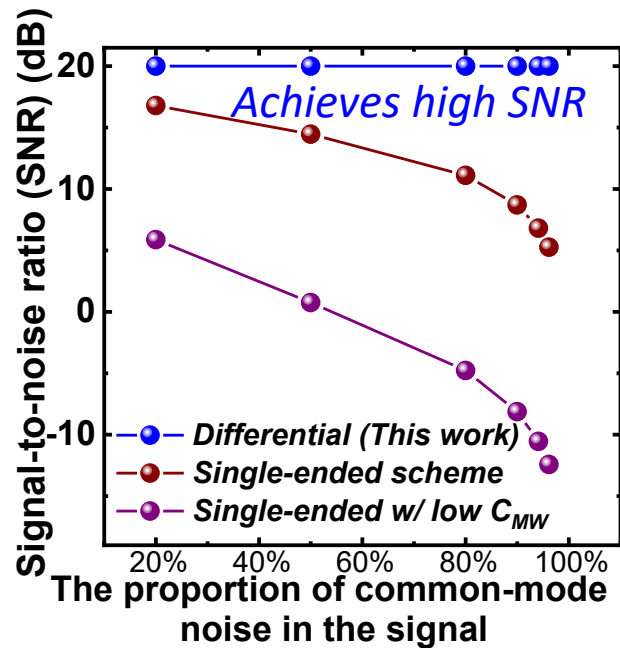
FS-Cap based Δ -CIM for LLM

- The FS-Cap based Δ -CIM for edge large language models (LLMs)
 - Unify highly parallel MAC and searching operation for LLMs
 - Constructed in HSPICE using FS-Cap device model



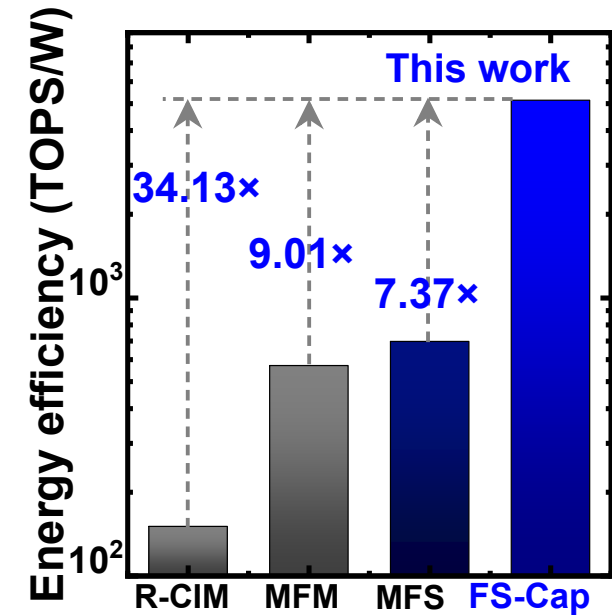
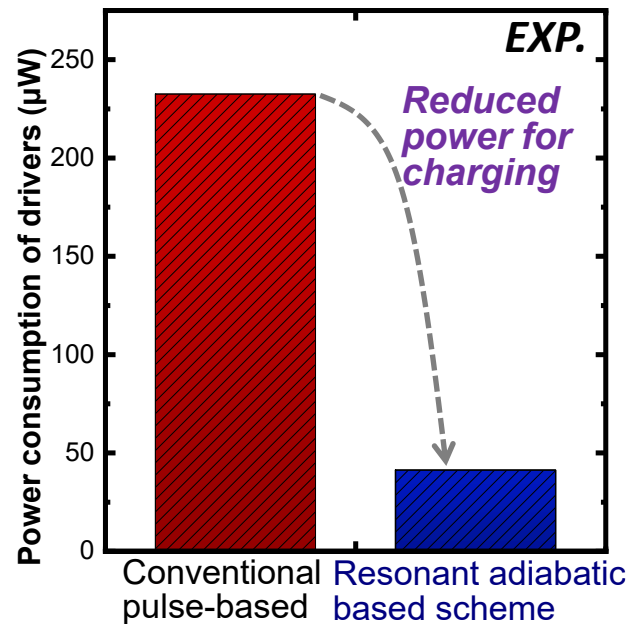
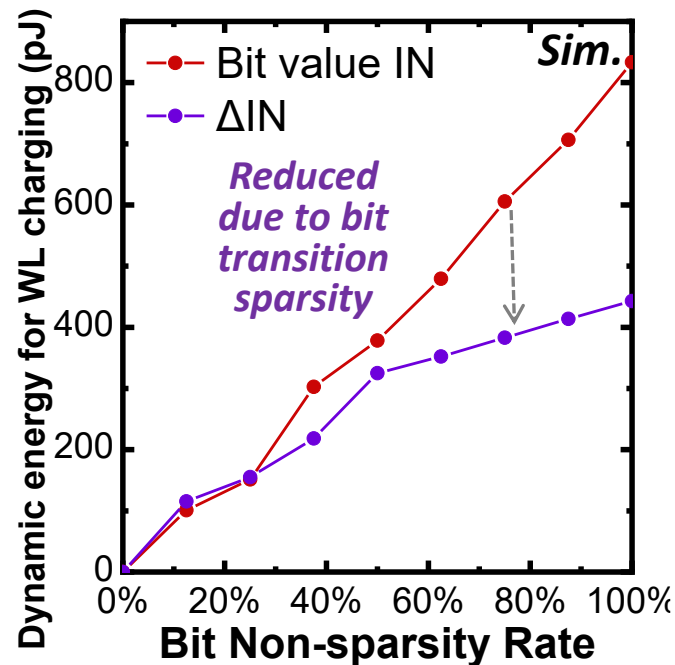
Performance evaluation of FS-Cap based Δ -CIM (1/2)

- Enhance signal-to-noise ratio (SNR) and parallel computing capability
 - Due to **three terminal FS-Cap with differential ports** in the proposed Δ -CIM
 - Guarantee accuracy in both inference and fine-tuning process



Performance evaluation of FS-Cap based Δ -CIM (2/2)

- Energy efficiency of the proposed FS-Cap based Δ -CIM
 - Dynamic power is reduced due to the **Δ IN scheme based self-adaptive sparsity**
 - Further improved by co-designing with our previous proposed adiabatic C-CIM



Benchmark of proposed FS-Cap based Δ -CIM with other reported memcapacitor-based CIM solutions

	IEDM23 [1]	IEDM24 [2]	IEDM24 [3]	Nat. Electron.[4]	VLSI22 [5]	<i>This work</i>
Structure	MFM	MANOS	MF-OS	MFS	MFS	<i>MIFIS</i>
Mechanism	Domain wall creep	Acc./Dep. of silicon	Acc./Dep. of silicon	Charge-shielding	Inv./Dep. of silicon	<i>Charge-steering</i>
Program pulse	3V, 1 μ s	14V, 1ms	6V, 1 μ s	6.1V, 1ms	6.5V, 2 μ s	<i>15V, 1μs</i>
MW (V_{TH} , V)	-	$\sim 2V$	$\sim 1V$	2.7V	1V	<i>6.3V</i>
MW (C, $\mu F/cm^2$)	0.14	0.17	2.63	0.004	0.25	<i>0.4</i>
On/off ratio	1.05	~ 1.5	1.67	1478	125	<i>51.2</i>
Read endurance	$\sim 10^{10}$	$> 10^6$	10^9	-	-	<i>$> 10^8$</i>
CIM scheme	Single-ended output w/ poor SNR and sparsity (two-terminal memcapacitor design)					<i>Δ-CIM (3-terminal)</i>
Subarray energy	51.7pJ ^(b)	-	-	95.6pJ	41.2pJ ^(b)	<i>0.65pJ^(b)</i>
Energy efficiency (TOPS/W) ^(c)	633.7 ^(b)	-	-	3452.6	794.1 ^(b)	<i>5140.6 TOPS/W^(b)</i>

(a) In the case that the common mode noise and differential mode noise are 3:1. (b) Calculated in HSPICE based on same array size with respectively proposed weight cell characteristics. (c) The energy of ADC is extracted from [6].

Outline

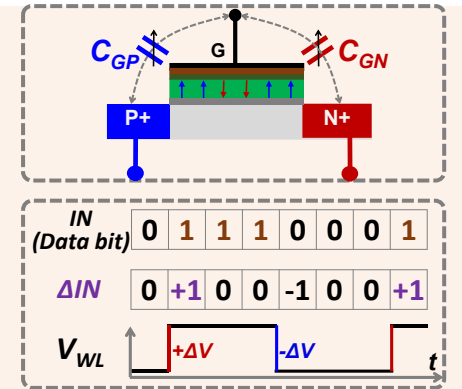
- FE charge-steering memcapacitor (FS-Cap) for Δ -CIM
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- FS-Cap based Δ -CIM macro for LLMs
- Summary

Summary

A novel FE charge-steering memcapacitor (**FS-Cap**) based **Δ -CIM** with **spatial & temporal delta-encoding**, enhancing parallelism & energy efficiency

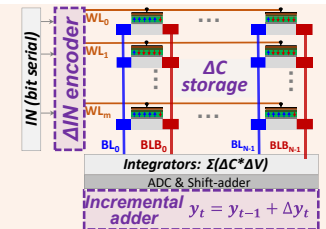
Co-design of FS-Cap device & Δ -compute architecture

- FS-Cap for spatial ΔC storage: **asymmetric doping** for dual-ended extraction of MOS cap. & FE-assisted gate-injection **steers charge between dual-ends**
- Temporal delta-input (ΔIN) encoding for FS-Cap array: **bit-transition-aware charging** $\rightarrow$ *adaptive sparsity and charging energy reduction*



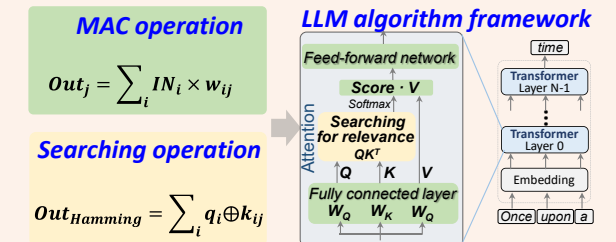
Experimental demonstration of FS-Cap-based Δ -CIM

- AND-type FS-Cap array based Δ -CIM system featuring noise suppression
- **Unified Δ -compute for MAC and searching** with reduced power consumption



FS-Cap-based Δ -CIM for edge LLM application

- Inference and fine-tuning with high signal-to-noise ratio (SNR)
- *High parallelism and energy efficiency* for edge AI



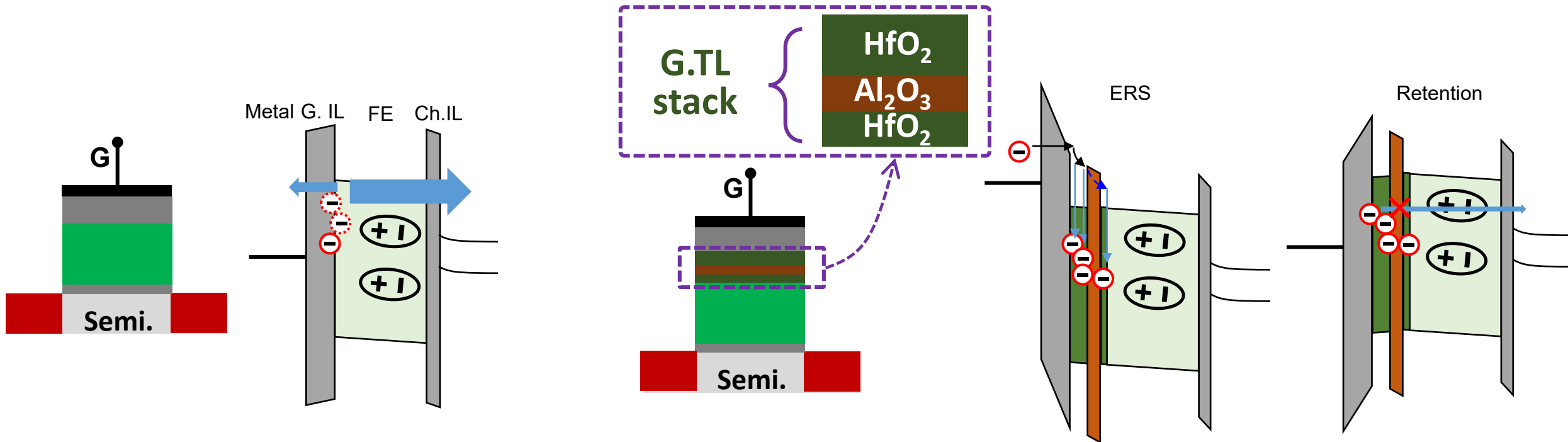
Thank you for your attention!

Backup

Gate-trapping layer design

—— Mechanism analysis

- Gate trapping layer: HfO_2 inserted with Al_2O_3 barrier layer



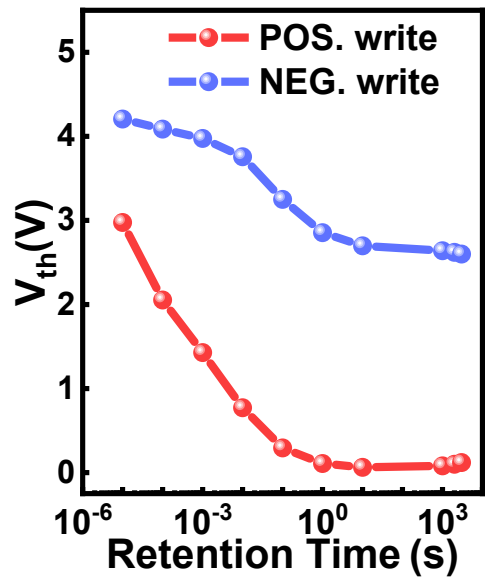
Conventional MIFIS structure

The proposed MIFIS structure with gate trapping layer stack

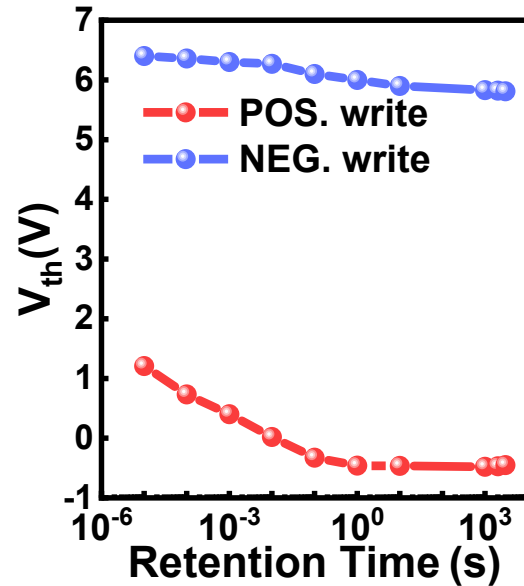
Gate-trapping layer design

——Performance improvement

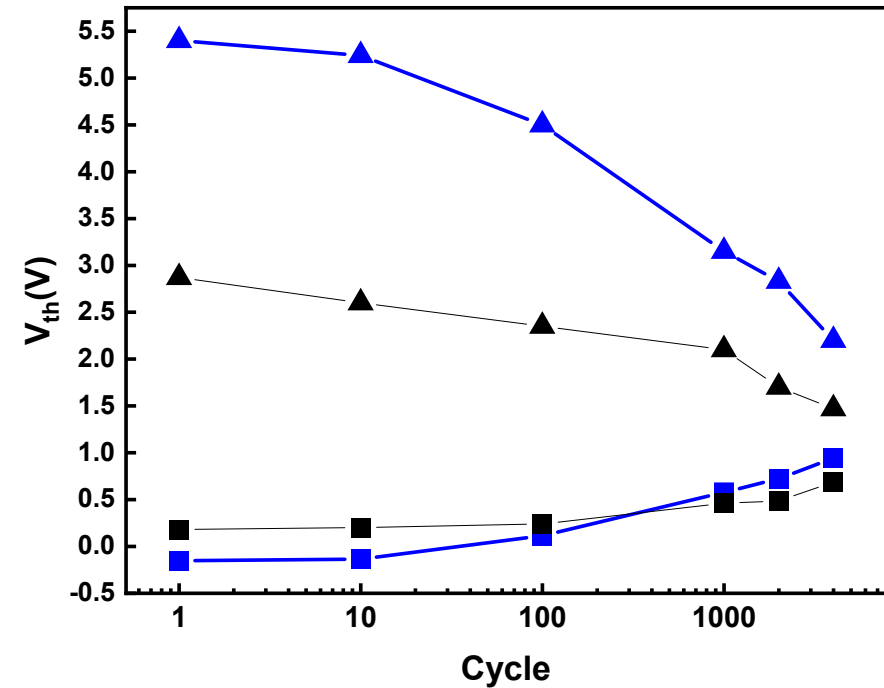
- Retention and endurance improvement



Conventional



The proposed



Unpublished data